RAJALAKSHMI ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF INFORMATION TECHNOLOGY
LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE
Thandalam, Chennai-602015

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Register No: 241001010

Year / Branch / Section: II - IT

Semester: IV

Academic Year: 2025-2026

INDEX

EXPNO: 1 Create Epic,Features, User Stories, Task	AZUREDEVOPS ENVIRONMENT SETUP

Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

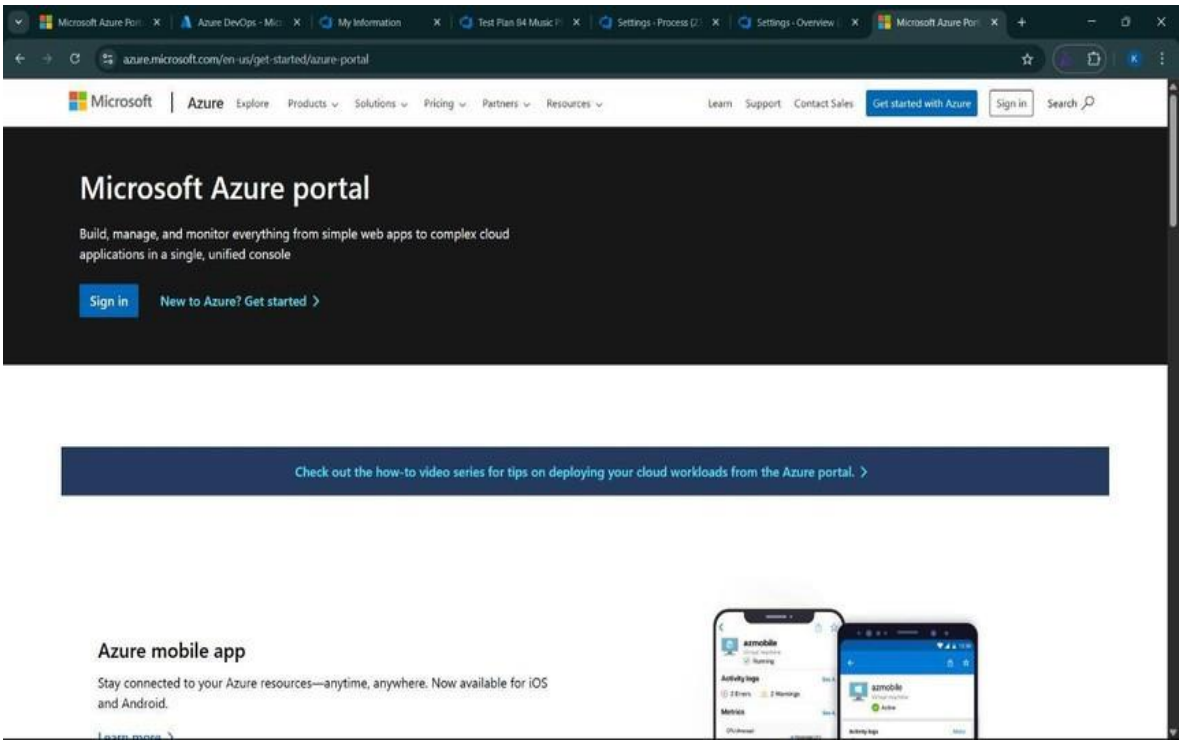
INSTALLATION:

1. Open your web browser and go to the Azure website:

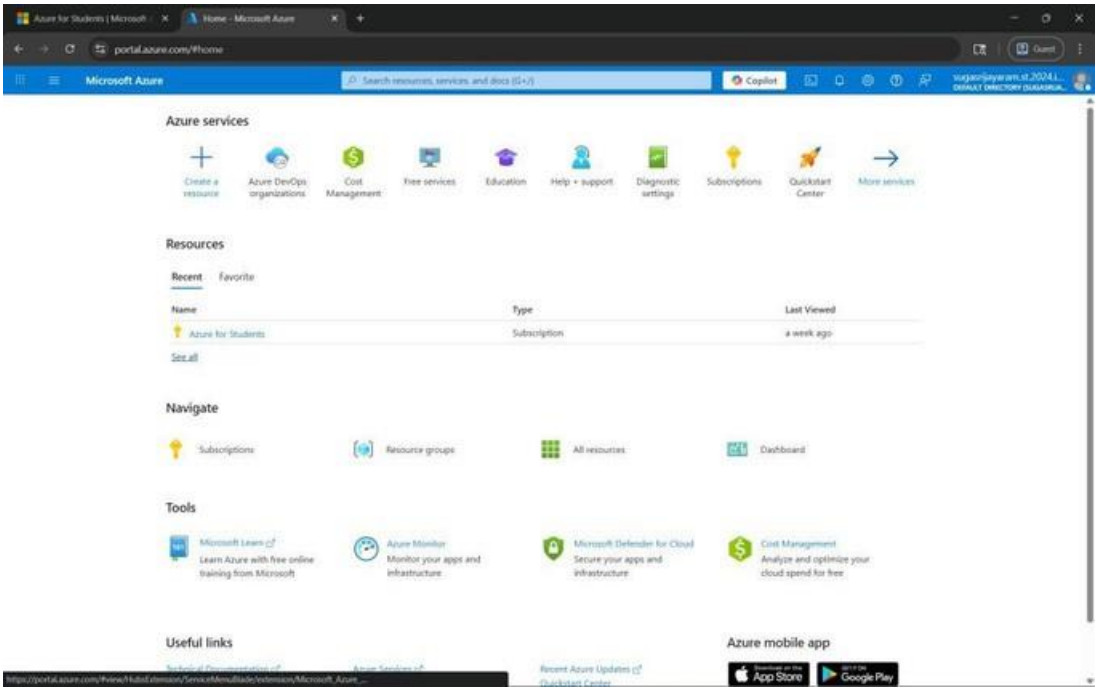
https://azure.microsoft.com/en-us/get-started/azure-portal.

Sign in using your Microsoft account credentials.

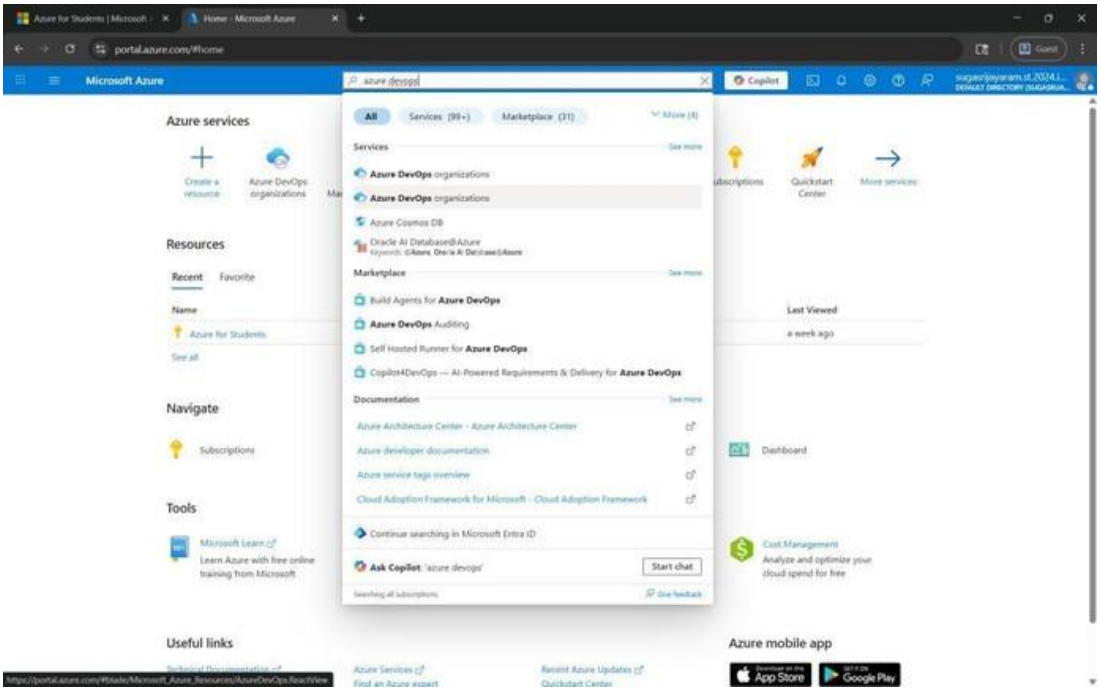
If you don't have a Microsoft account, you can create one here: https://signup.live.com/?lic=1



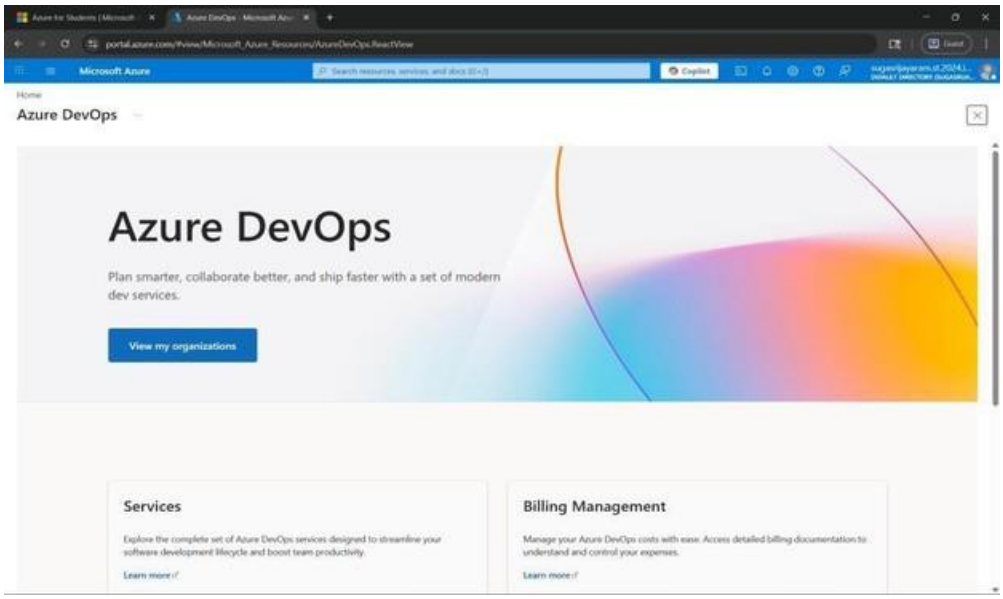
2. Azure home page

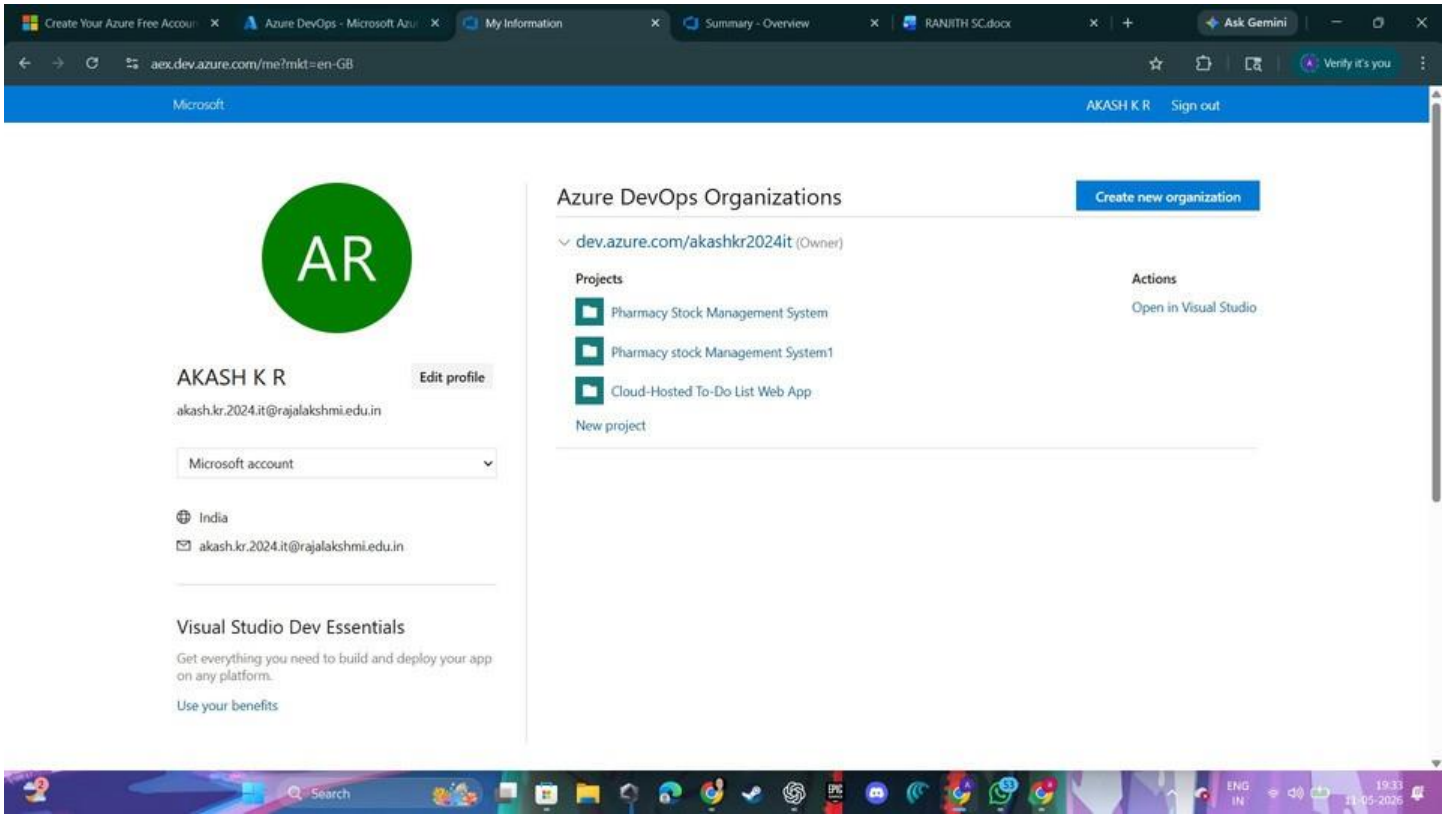


3. Open DevOps environment in the Azure platform by typing***Azure DevOps Organizations*** in the search bar.



4. Click on the ***View My Organization*** link and create an organization and you should be taken to the Azure DevOps Organization Home page.





Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

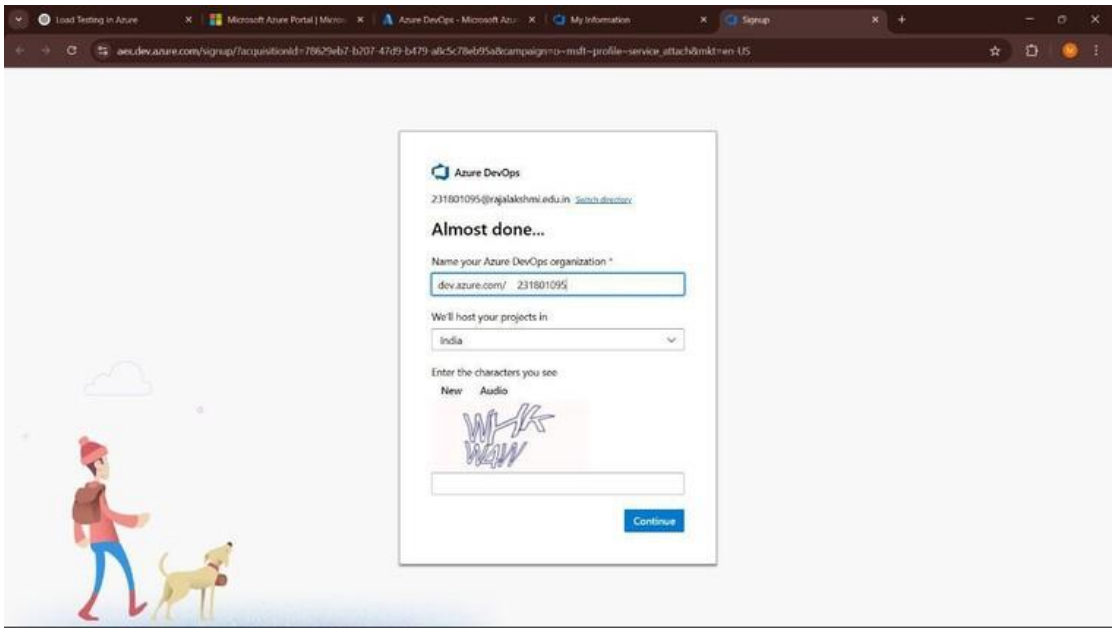
EXP NO: 2 Create Epic, Features, User Stories, Task	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

Procedure:

1. Create An Azure Account



2. Create the First Project in Your Organization

a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's **Home page**, click on the **New Project** button.

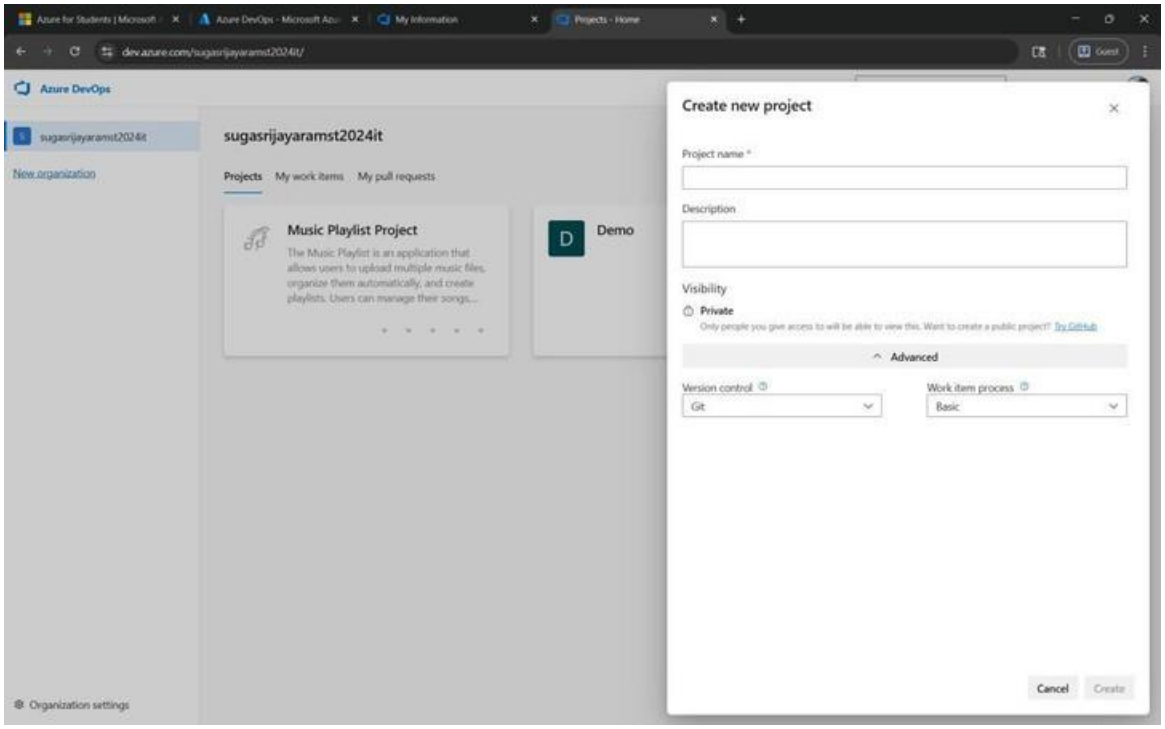
c. Enter the project name, description, and visibility options:

***Name**:* **Music Playlist Project**.

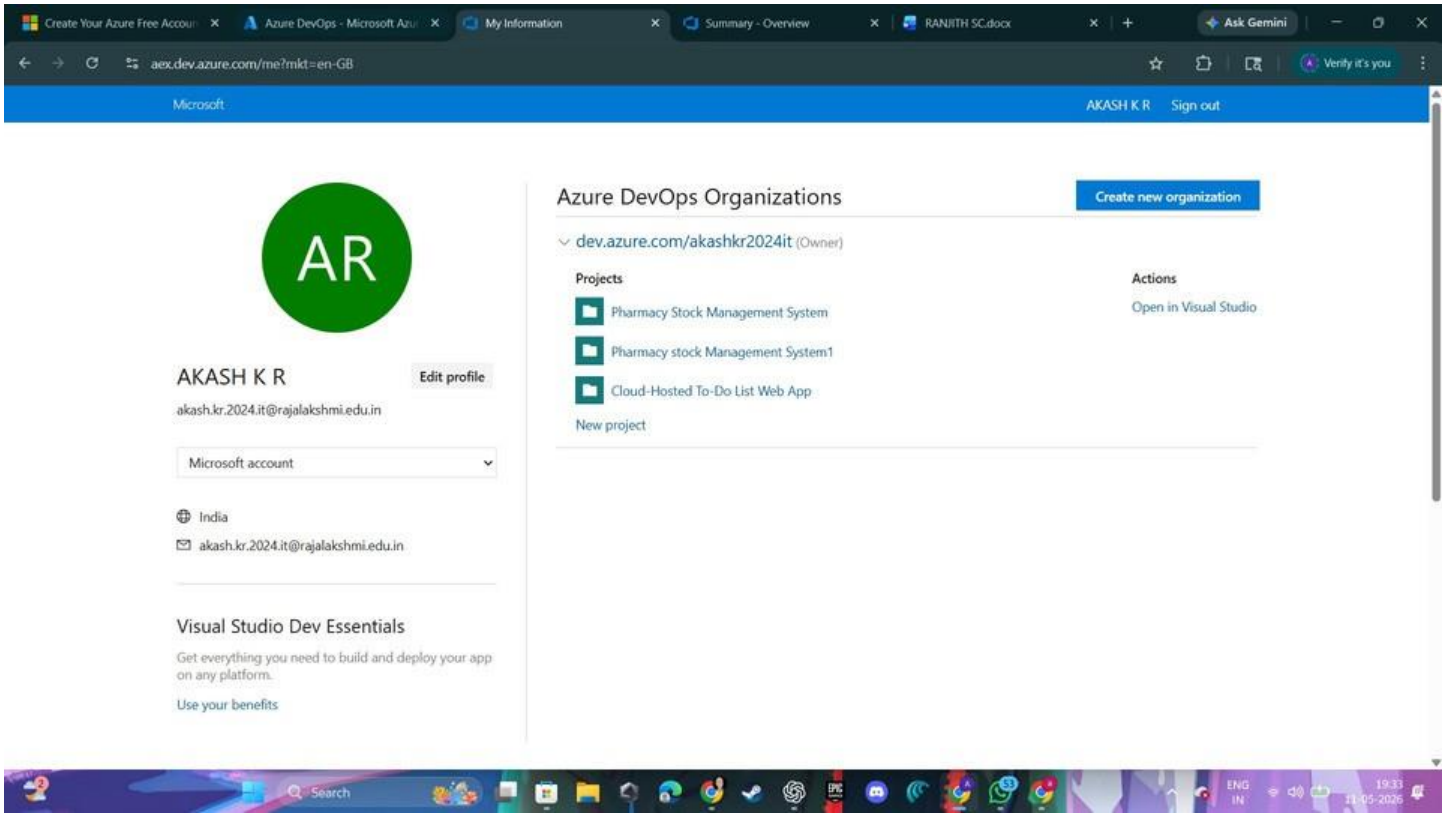
***Description**:* The Music Playlist is an application that allows users to upload multiple music files, organize them automatically, and create playlists. Users can manage their songs, arrange playlists, and download songs securely after payment. The system also uses external APIs to fetch song metadata and improve music organization.

***Visibility**:* Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

d. Once you've filled out the details, click **Create** to set up your first project.



3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.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Azure DevOps - Microsoft Azu
My Information
Summary - Overview
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Ask Gemini
aex.dev.azure.com/me?mkt=en-GB
Verify it's you
Microsoft
AKASH K R
Sign out
AR
AKASH K R
Edit profile
akash.kr.2024.it@rajalakshmi.edu.in
Microsoft account
India
akash.kr.2024.it@rajalakshmi.edu.in
Visual Studio Dev Essentials
Get everything you need to build and deploy your app on any platform.
Use your benefits
Azure DevOps Organizations
Create new organization
dev.azure.com/akashkr2024it (Owner)
Projects
Pharmacy Stock Management System
Pharmacy stock Management System1
Cloud-Hosted To-Do List Web App
New project
Actions
Open in Visual Studio
Search
ENG
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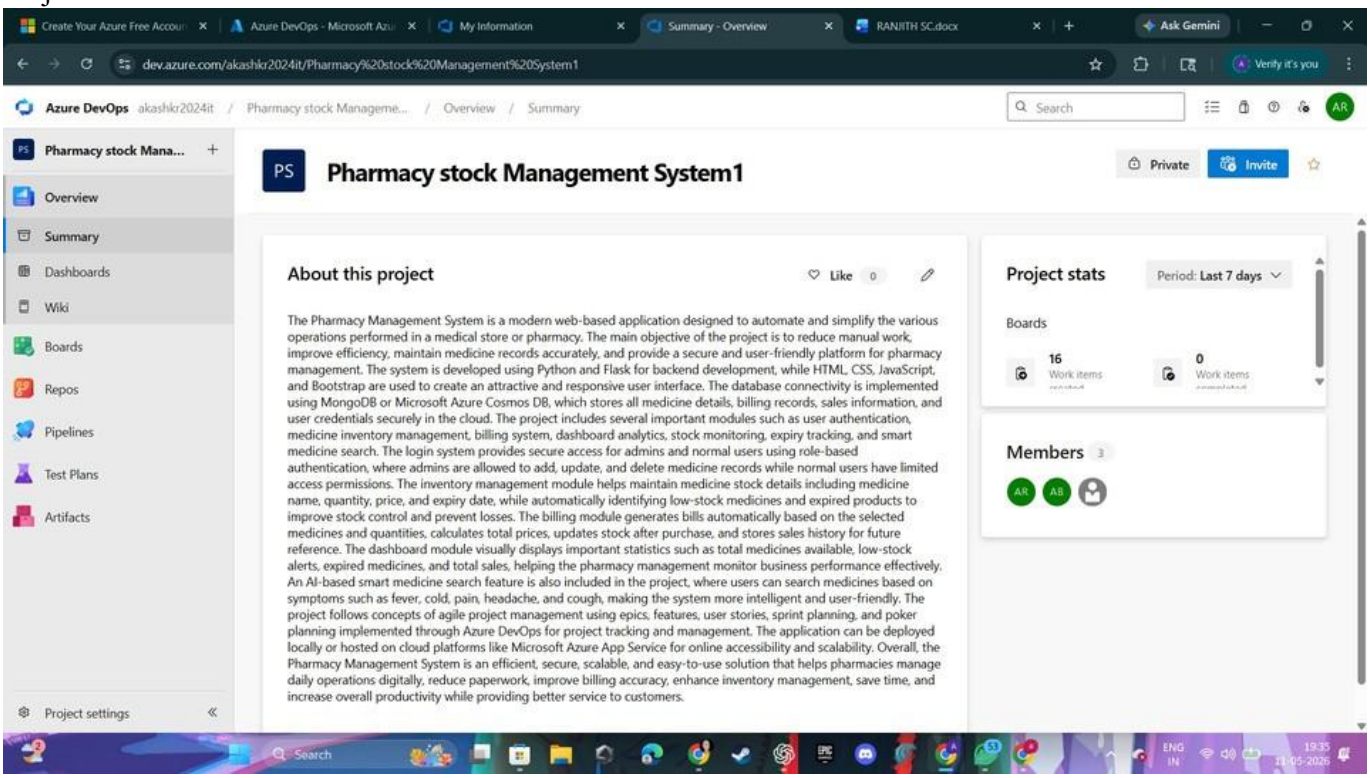
11-05-2026

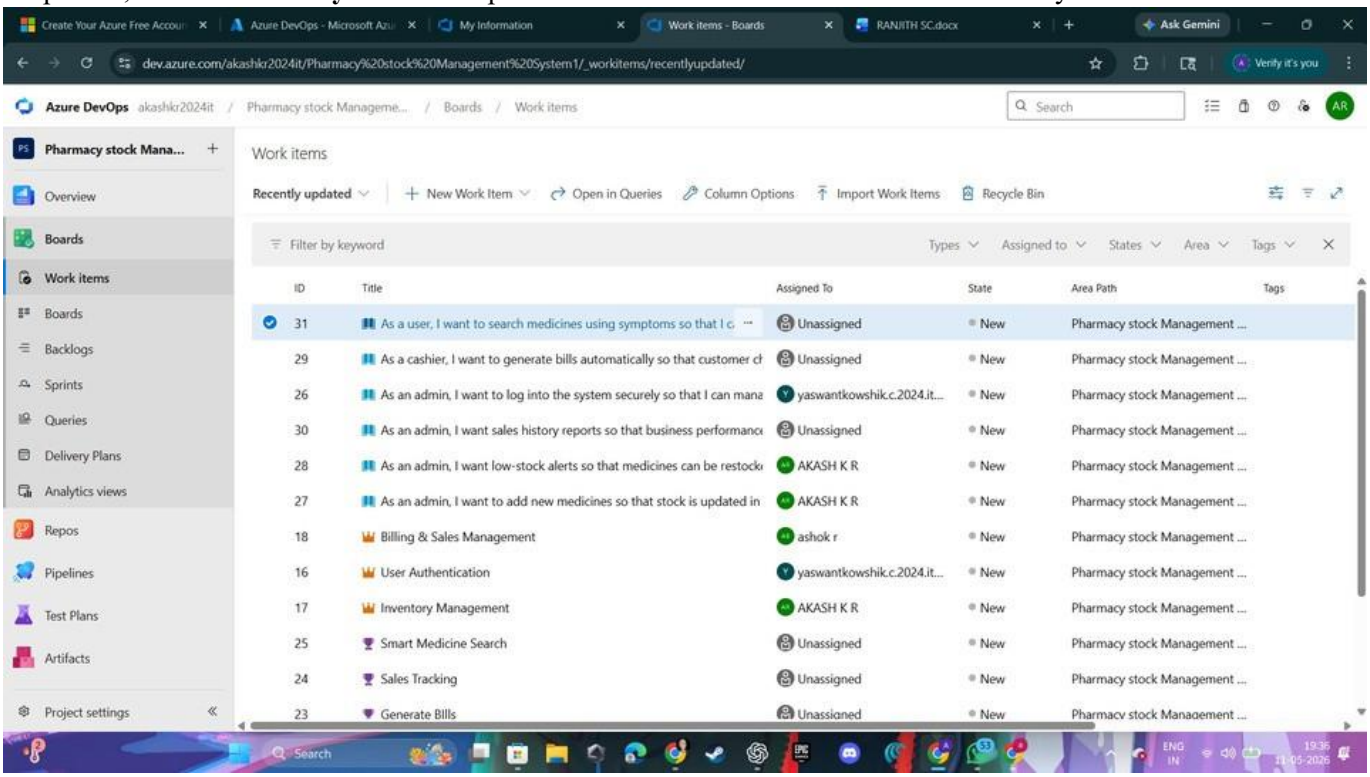
4. Project dashboard

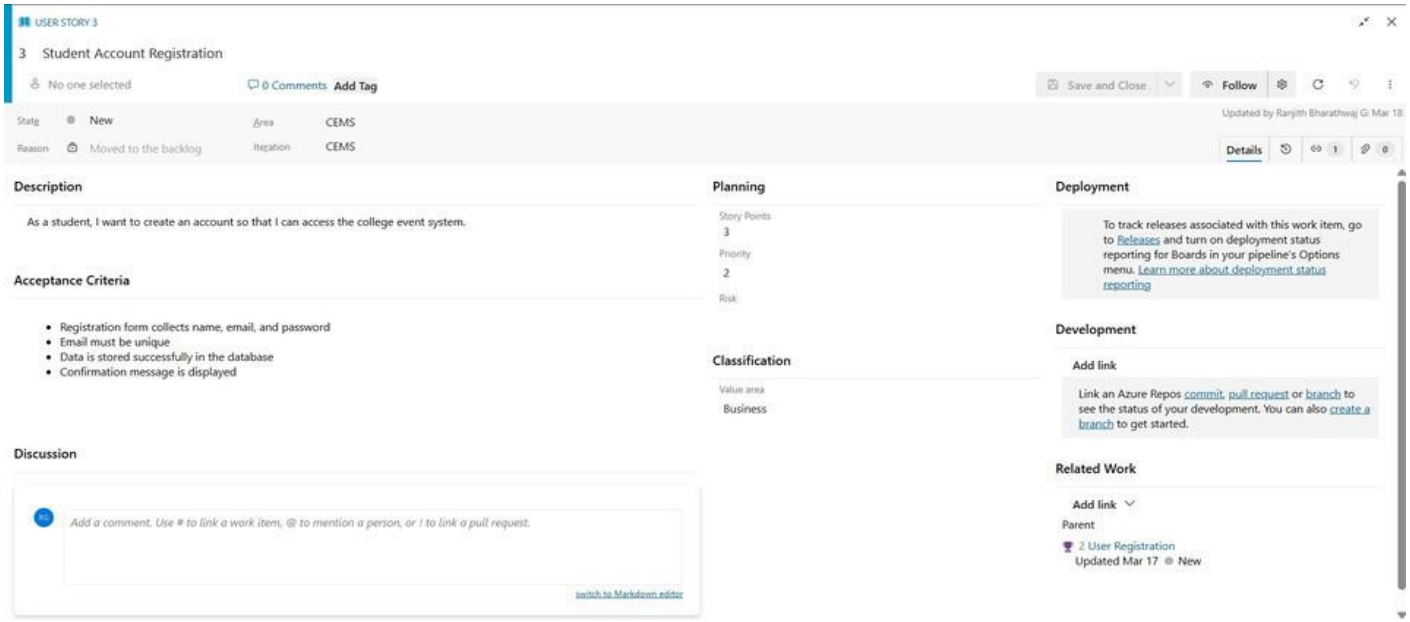


5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Addaworkitem** dropdown, select **User Story**. This will open a form to enter details for the new User Story.





Result:

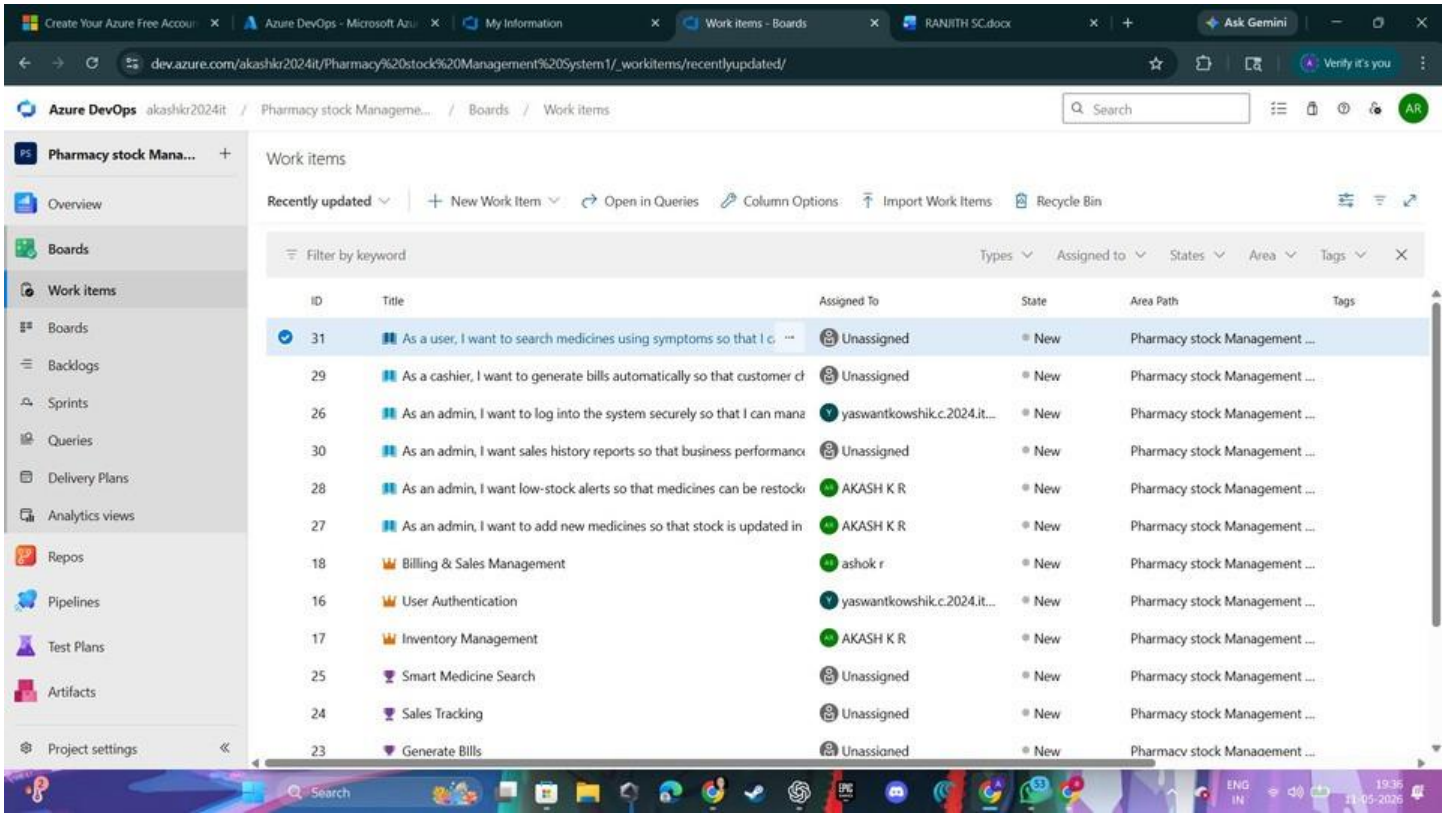
Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3 Create Epic, Features, User Stories, Task	SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

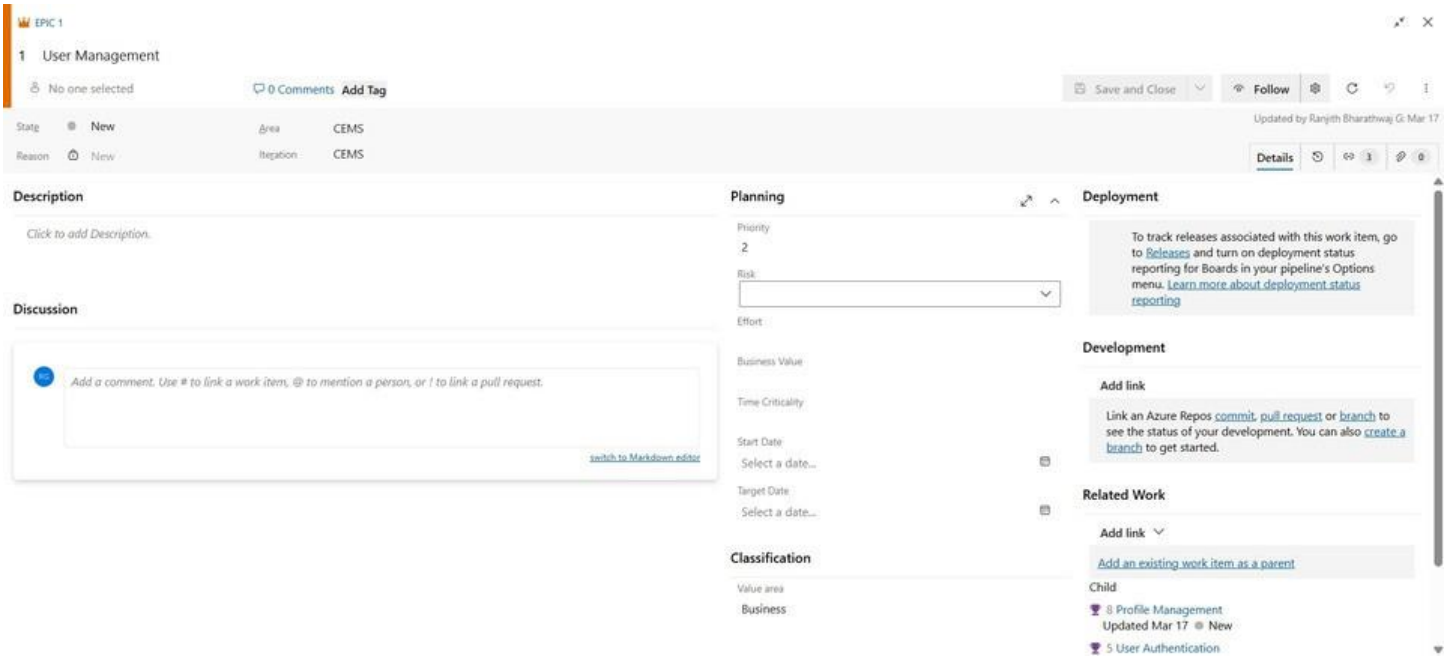
Aim:

To learn about how to create epics, user story, features, backlogs for college event management.

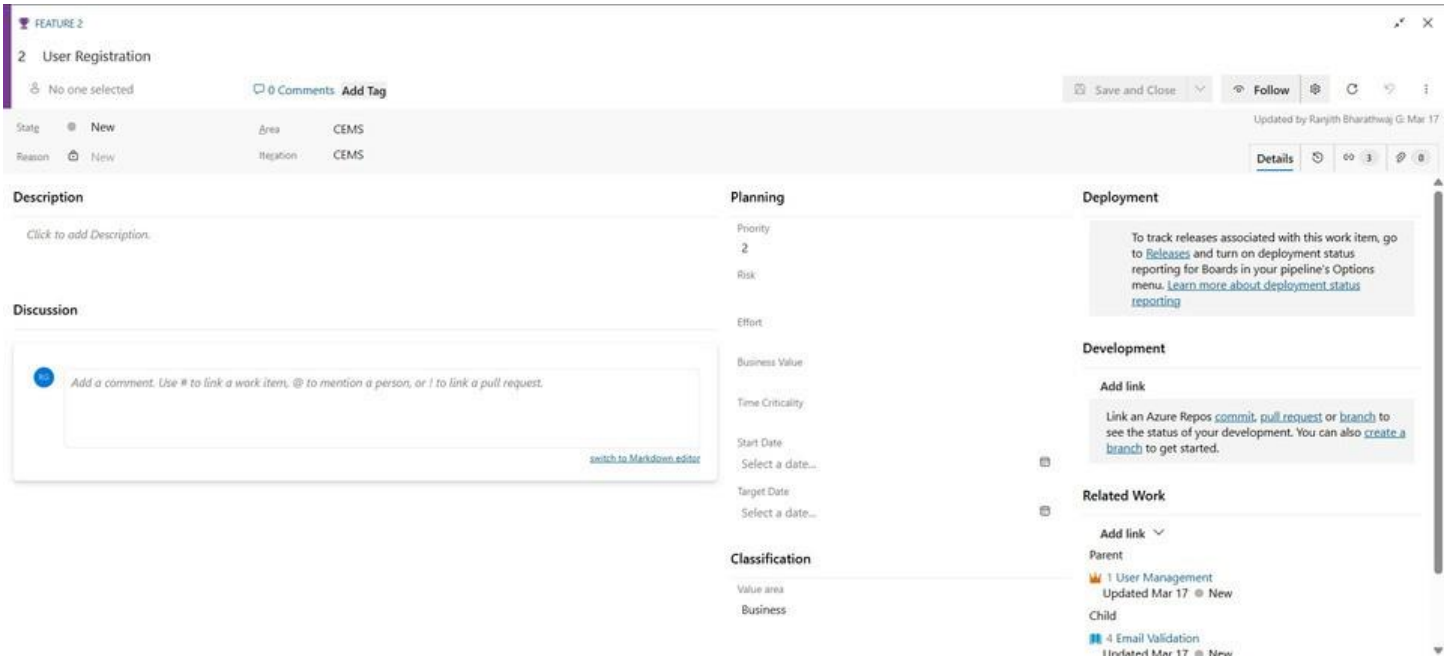
Create Epic, Features, User Stories, Task:



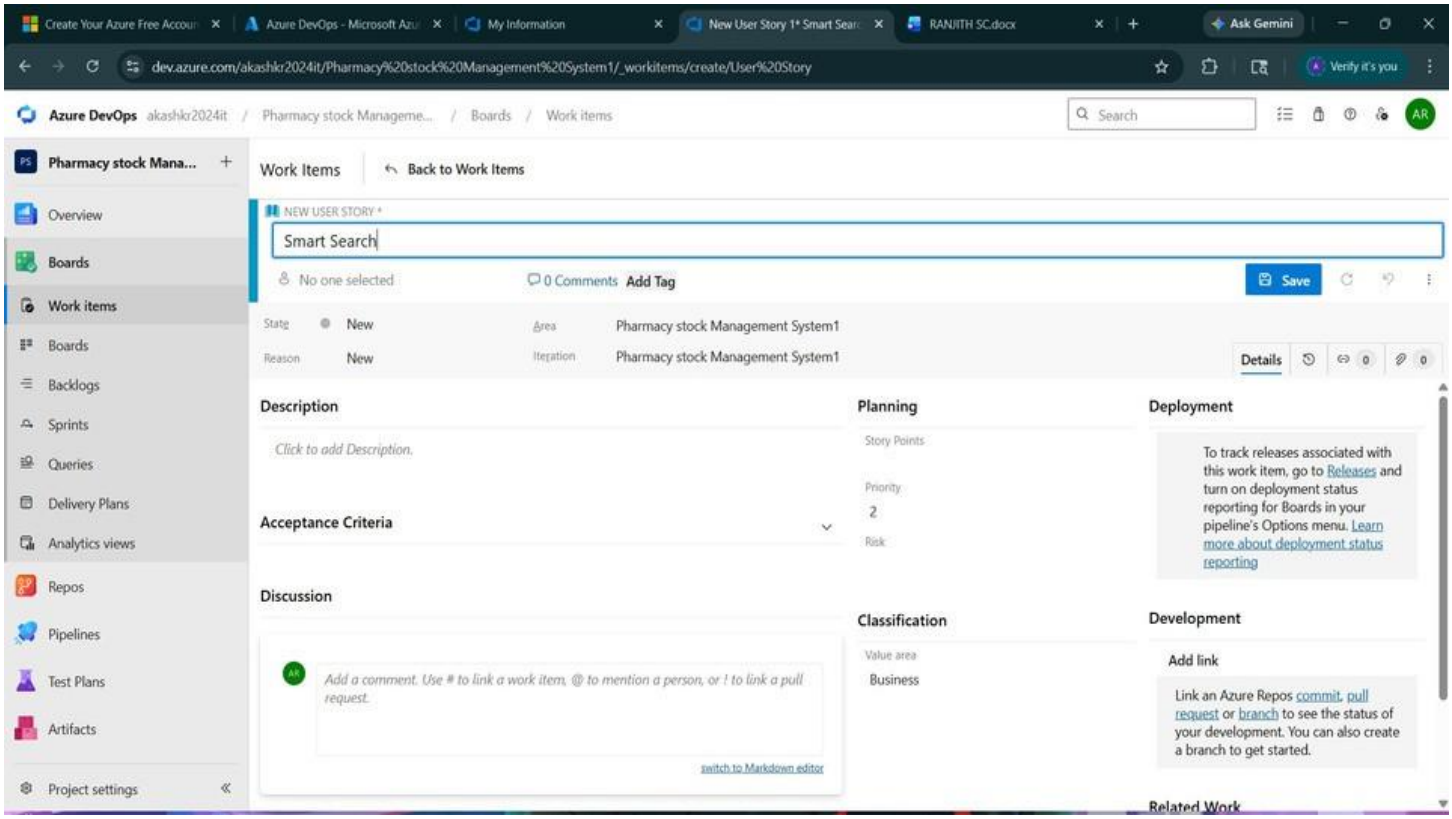
Fill in Epics



Fill in Features



Fill in User Story Detail



Result:

Thus, the creation of epics, features, user story and task has been created successfully.

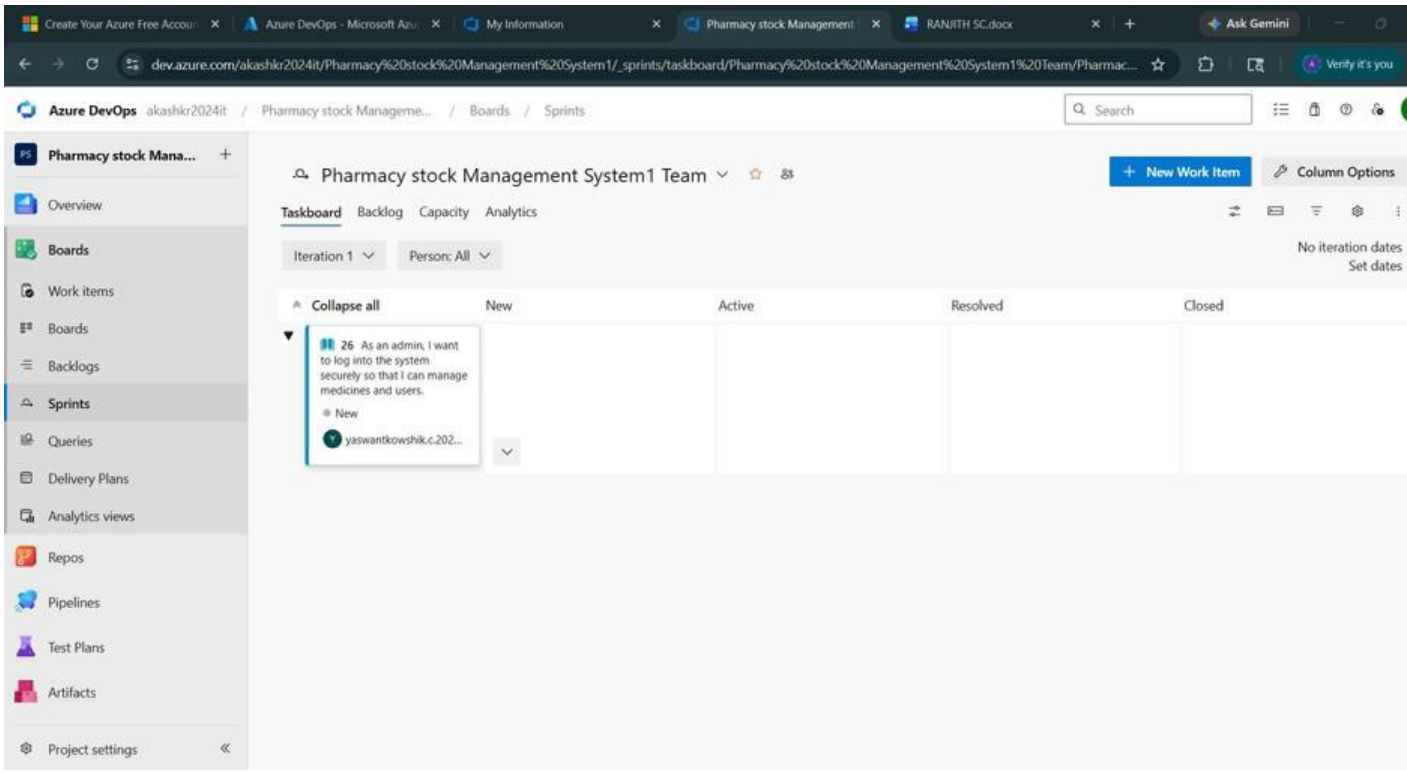
EXPNO: 4 Create Epic,Features, User Stories, Task	SPRINT PLANNING

Aim:

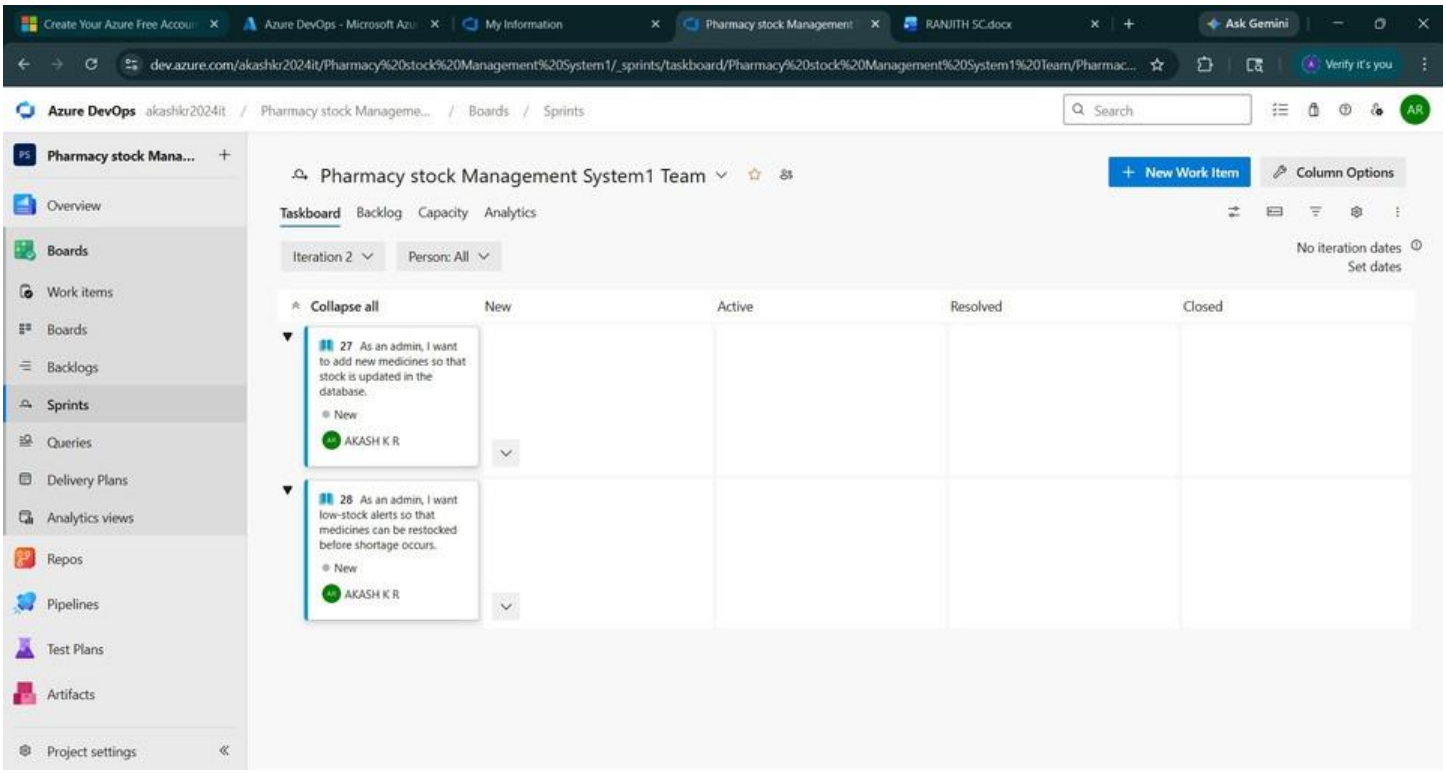
To assign user story to specific sprint for the College event management Project.

Sprint Planning:

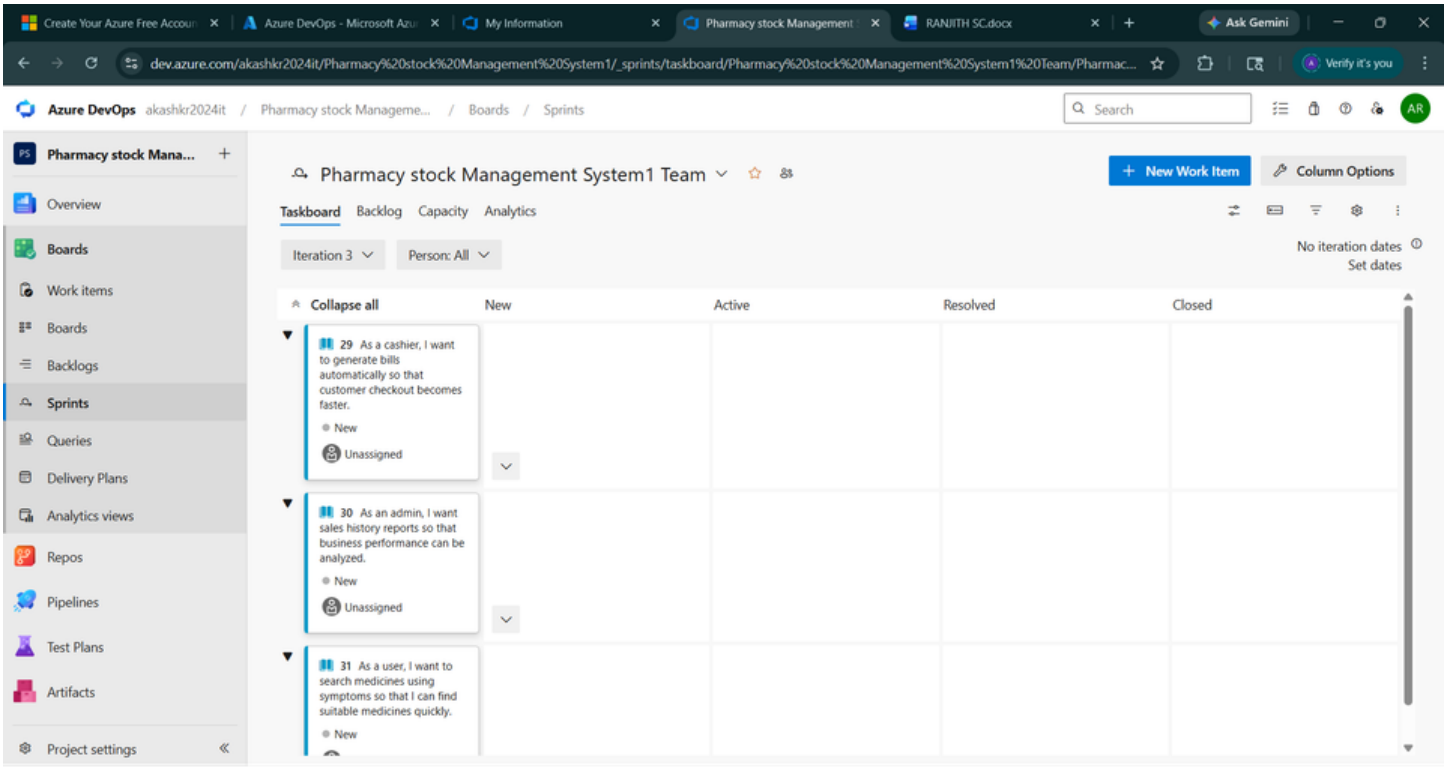
Sprint 1



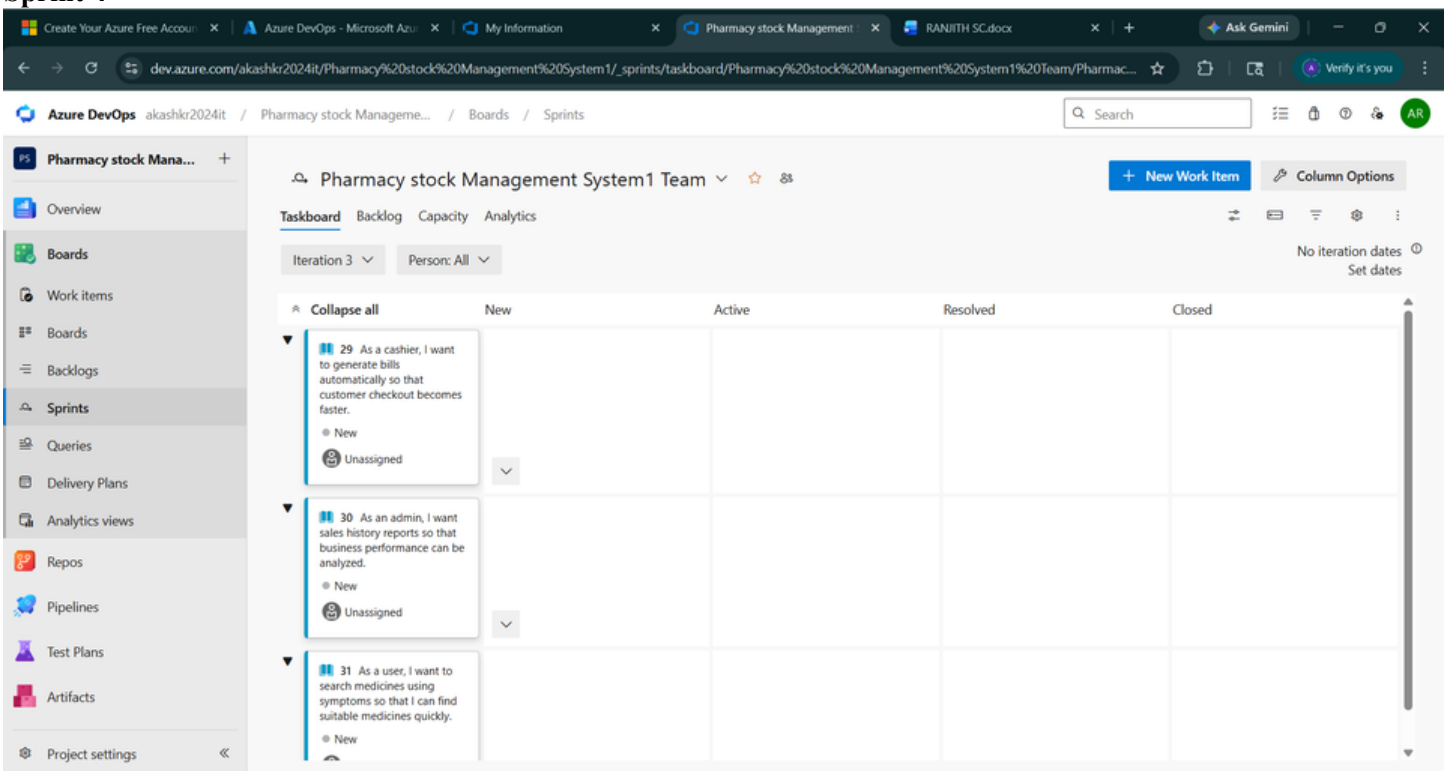
Sprint 2

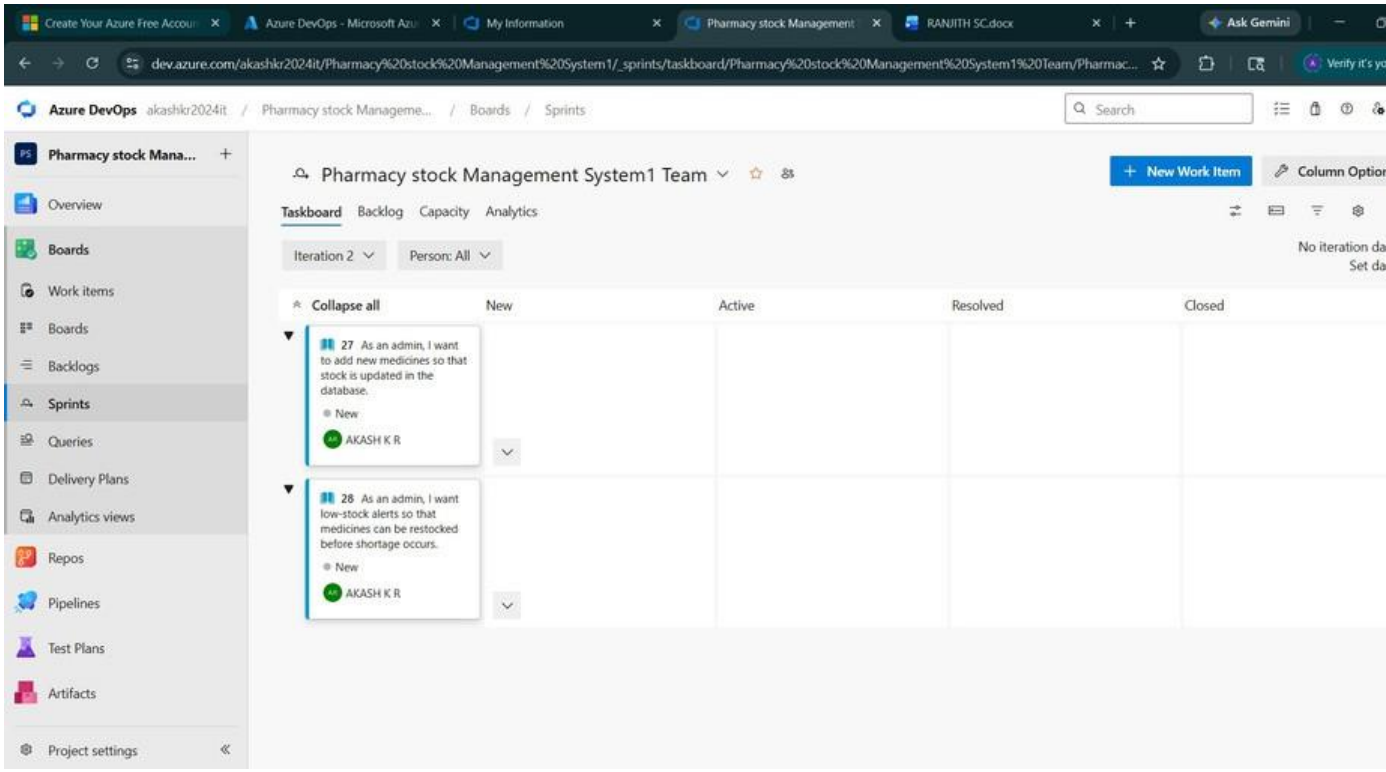

Azure DevOps
Pharmacy stock Management System1 Team
New Work Item
Column Options
Taskboard
Backlog
Capacity
Analytics
Iteration 2
Person: All
No iteration dates
Set dates
Collapse all
New
Active
Resolved
Closed
27 As an admin, I want to add new medicines so that stock is updated in the database.
New
AKASH K R
28 As an admin, I want low-stock alerts so that medicines can be restocked before shortage occurs.
New
AKASH K R
Overview
Boards
Work items
Boards
Backlogs
Sprints
Queries
Delivery Plans
Analytics views
Repos
Pipelines
Test Plans
Artifacts
Project settings

Sprint 3



Sprint 4





Result:

The Sprints are created for the College event management.

EXPNO: 5 Create Epic,Features, User Stories, Task	POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories for Music Playlist Project.

Planning Poker Estimation:

User Story ID: 20

Title: User Registration and Authentication

Description:

As a user, I want to register and log in securely so that I can access my personal music playlists and downloads.

Story Point Estimation: 8 Points

Team Poker Votes:

Team Member	Vote (Story Points)
Team Member 1	5
Team Member 2	8
Team Member 3	8
Team Member 4	8

Final Consensus: 8 Points

Reason for Estimation:

- Includes multiple features: registration, login, session, logout
- Requires frontend + backend + database integration
- Security implementation (password hashing, authentication)
- Moderate complexity with high risk

Risks:

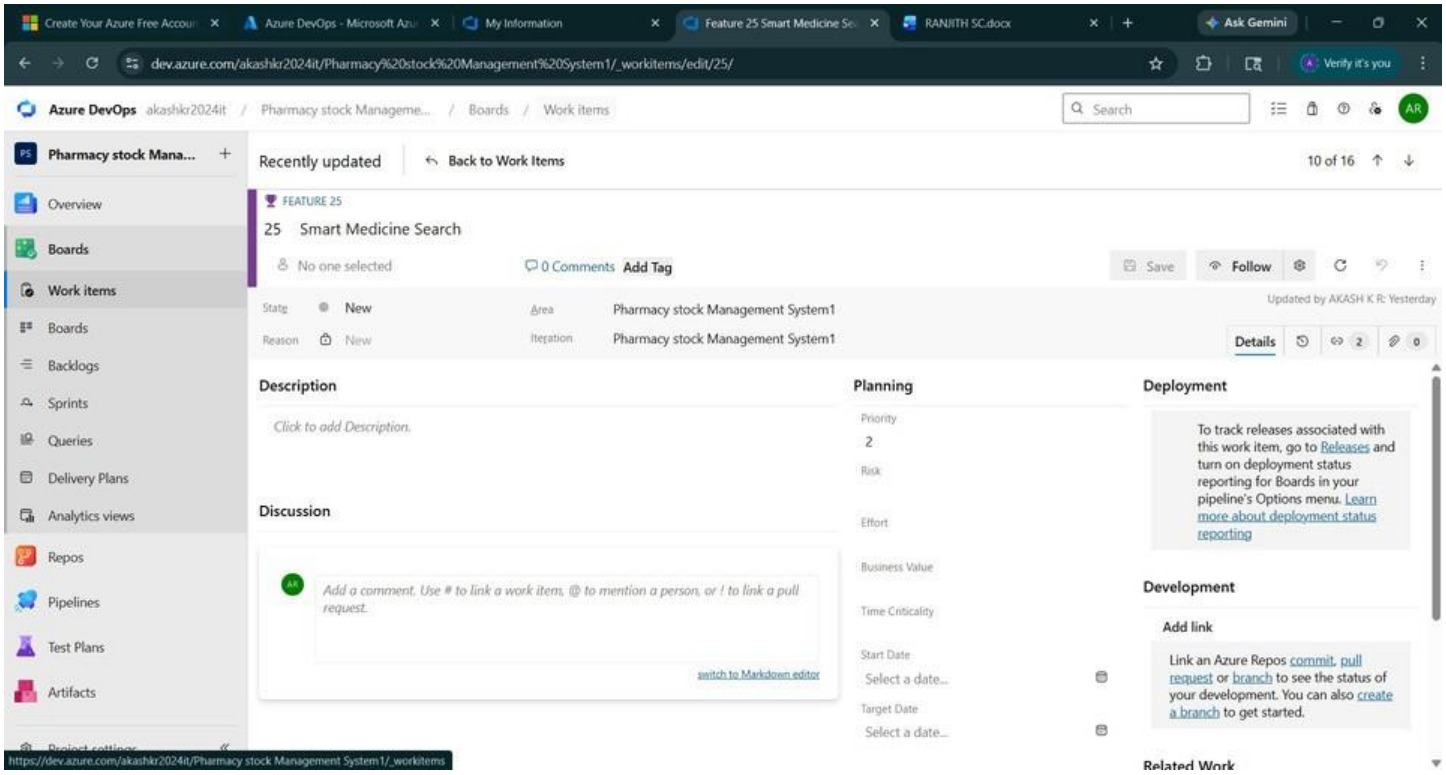
- Security vulnerabilities
- Session management issues
- Integration challenges

Acceptance Criteria Covered:

- User registration with email & password
- Login with valid credentials
- Error message for invalid login
- Session maintained after login
- Logout functionality

Conclusion:

Finalagreed estimation is **8 story points** based on complexity, effort, and risk.



Result:

The Estimation/Story Points is created for the project using Poker Estimation.

EXPNO: 6 Create Epic,Features, User Stories, Task	DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

Aim:

To Design a Class Diagram and Sequence Diagram for the College Event Management.

Class Diagram:

Classes:

- Student
- Admin
- Event
- Registration
- Payment
- Certificate
- Notification
- Coordinator

Key Attributes:

Student:

StudentId,Name,Email,Department

Event:

EventId,Title,Date,Venue,Category,Fee

Registration:

RegistrationId,StudentId,E

ventId,PaymentStatus

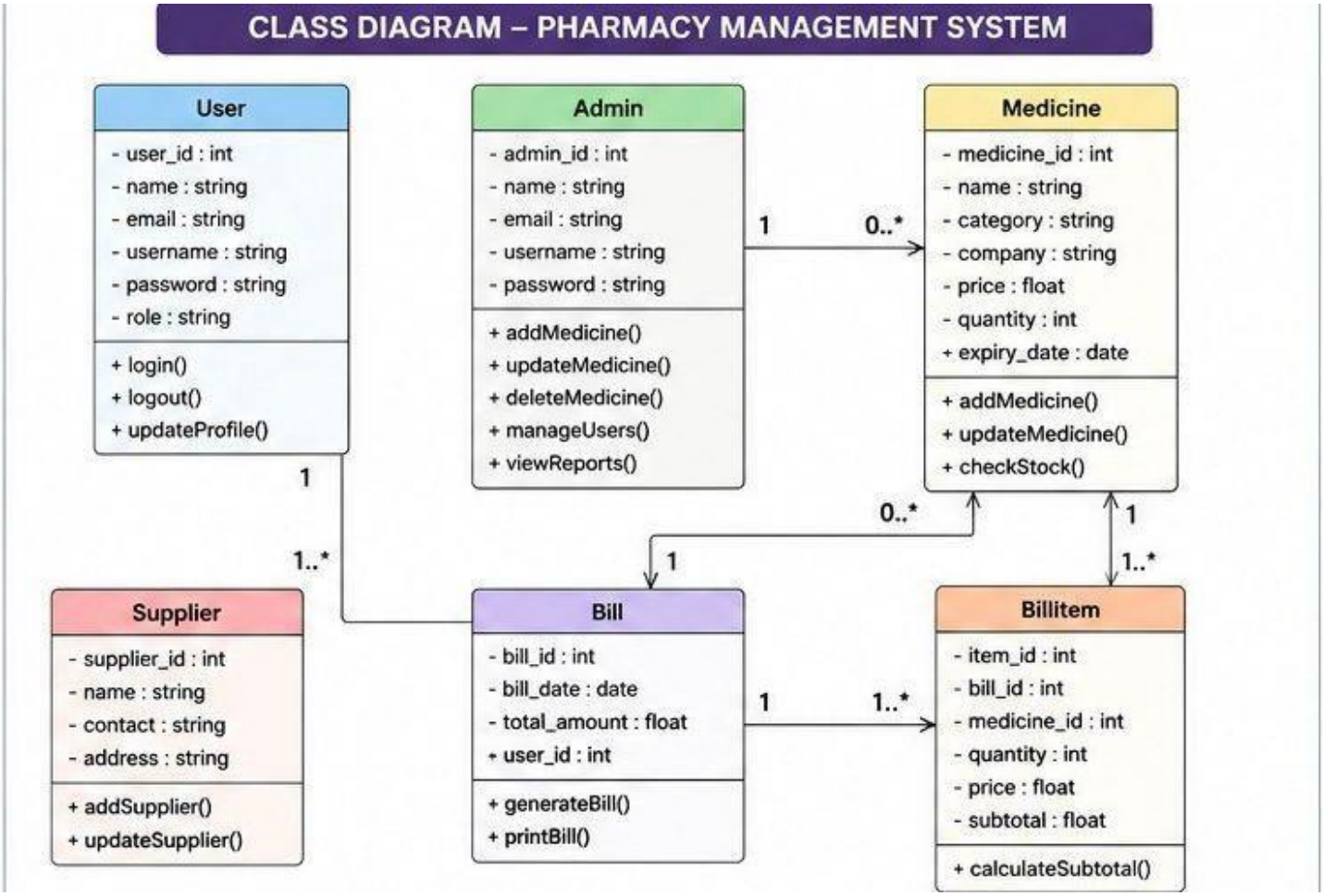
Payment:

PaymentId,Amount,Tran

sactionStatus

Relationships:

- Student → registers for → Event
- Activity → conducted at → Venue
- Student → participates in → Competition
- Faculty Coordinator → manages → Event
- Organizer → schedules → Event
- Event → has → Participants
- Student → makes → Payment
- Student → receives → Ticket/Pass
- Student → submits → Feedback
- Student → receives → Notification
- Sponsor → sponsors → Event
- Event → awards → Certificate/Prize
- Department → organizes → Event
- Venue → allocated to → Event
- Student → joins → Team
- Team → participates in → Event
- Admin → approves → Event Request
- Event → publishes → Results
- Offer/Discount → applies to → Event Registration



Sequence Diagram:

Actors:

User→Frontend → Backend → Database → Auth Service

Steps:

1. Userentersemail and password
2. Frontend validates input (email format, password)
3. Frontend sends login request to backend
4. Backend validates request data
5. Backend checks user in database

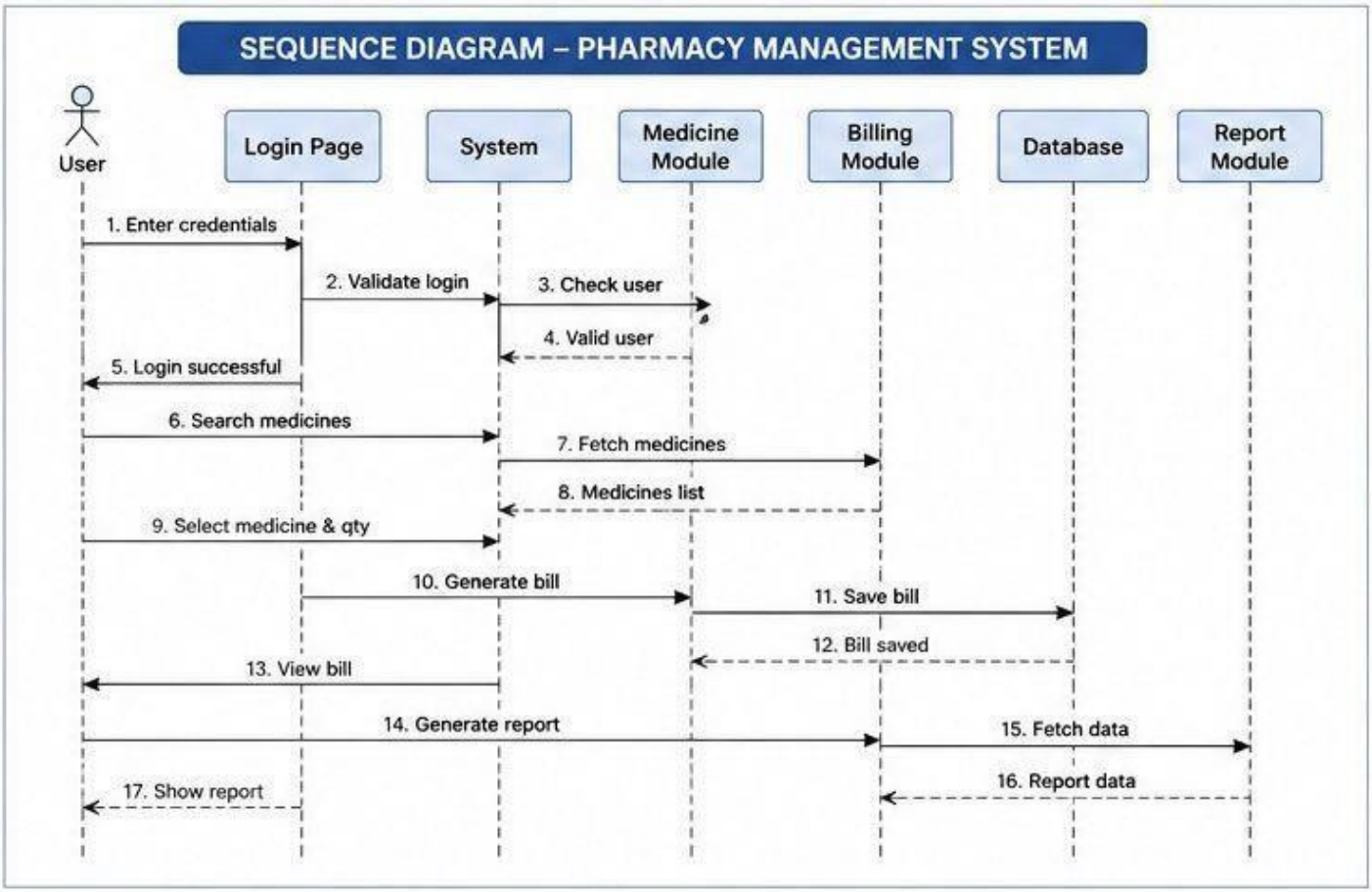
Condition: User Not Found

- Backendreturns error: *Invalid credentials*

Condition: User Found

6. Backendverifiespassword
7. If password incorrect → return error
8. If password correct → proceed
9. Backend verifies user with auth service
10. Backend generates JWT token
11. Backend sets session (token expiry)
12. Backend sends success response

13. Frontend stores token
14. Frontend redirects to dashboard
15. User login successful



Result: The Class Diagram and Sequence Diagram is designed Successfully for the College Event Managemnt.

EXPNO: 7 Create Epic,Features, User Stories, Task	DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

Architectural Diagram:

Layers:

1. Client Layer

- Browser
- React App
- Zustand (State Management)

2. API Communication

- Axios (HTTP Requests)

3. Backend Layer

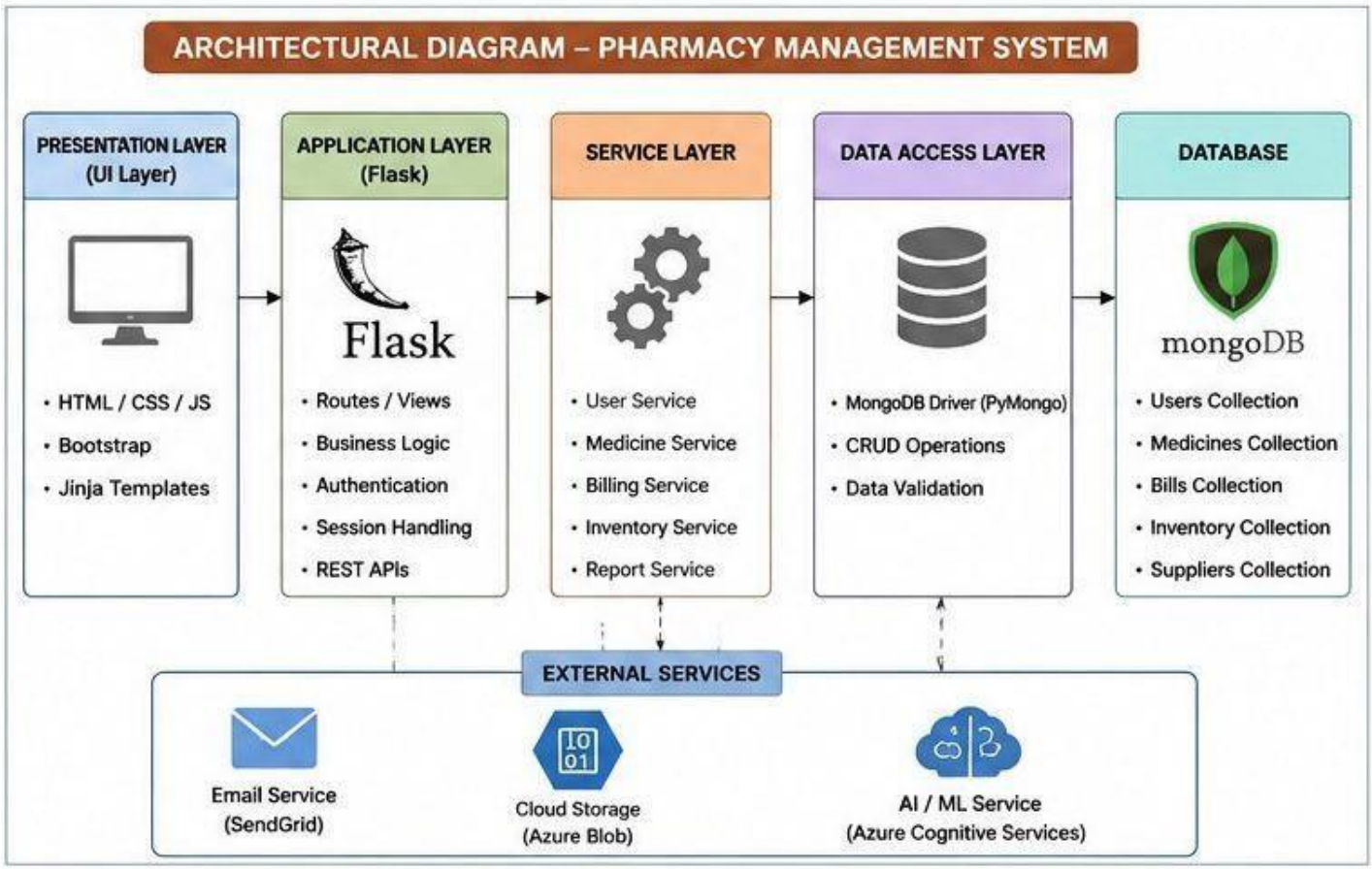
- Express Server
- Routes
- Controllers

4. Services

- Clerk Authentication
- File Upload(Cloudinary)

5. Database Layer

- MongooseModels
- MongoDB



ER Diagram:

Entities & Attributes:

Student

- StudentId
- Name Email
- Department
-

Event:

- EventId
- Title
- Venue

- Date
- Fee

Registration

- registration
- amount
- status

Payment

- paymentId
- amount
- status

---

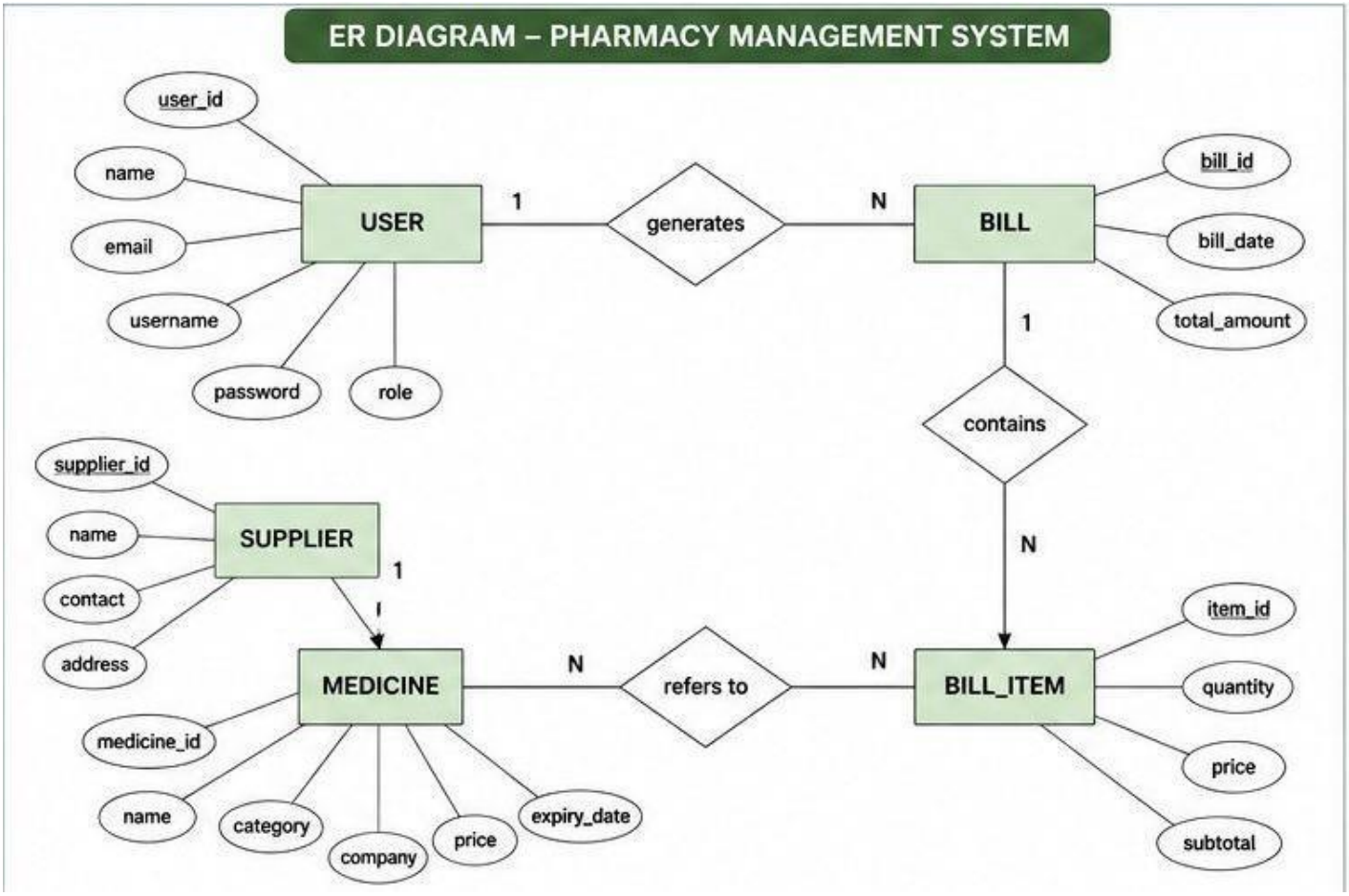
certificate

- certificateId

- issueDate

Relationships

- Student registers for Event (M : N)
- Event includes Activity (1 : M)
- Activity conducted at Venue (M : 1)
- Student participates in Competition (M : N)
- Faculty Coordinator manages Event (1 : M)
- Organizer schedules Event (1 : M)
- Event has Participants (1 : M)
- Student makes Payment (1 : M)
- Student receives Ticket / Pass (1 : 1)
- Student submits Feedback (1 : M)
- Student receives Notification (1 : M)
- Sponsor sponsors Event (M : N)
- Event awards Certificate / Prize (1 : M)
- Department organizes Event (1 : M)
- Venue allocated to Event (1 : M)
- Student joins Team (M : N)
- Team participates in Event (M : N)
- Admin approves Event Request (1 : M)
- Event publishes Results (1 : 1)
- Offer / Discount applies to Event Registration (1 : M)



Result: The Architecture Diagram and ER Diagram is designed Successfully for the College Event Management.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
Create Epic, Features, User Stories, Task	

Aim:

Test Plans and Test Case and write two test cases for at least two user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case:
Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Viewing and Managing Playlists
- Fetching Real-time Metadata
- Editing playlists (rename, reorder, record)
- Creating smart audio playlists based on categories (mood, genre, artist, etc.)

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).

3. Design Happy Path Test Cases

- Focused on validatingthat all features function as expected under normal conditions.
- Example: User logs insuccessfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test robustness and error handling.
- Example: Login fails with invalid credentials, save fails when offline, no recommendations found.

5. Break Down Steps and Expected Results

- Each test case contains step-by-stepactions and a corresponding expected outcome.
- Ensures clarity for both testers andautomation scripts.

6. Use Clear Naming and IDs

- Test cases are namedclearly (e.g., TC01 – Successful signup, TC06 – Add and Remove Songs from Playlist).
- Helps in quick identification and linking to user stories or features.

7. Separate Test Suites

- Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).
- Improves organization and test execution flow in Azure DevOps.

8. Prioritize and Review

- Critical useractions are marked high-priority.
- Reviewed forcompleteness and traceability against feature requirements.

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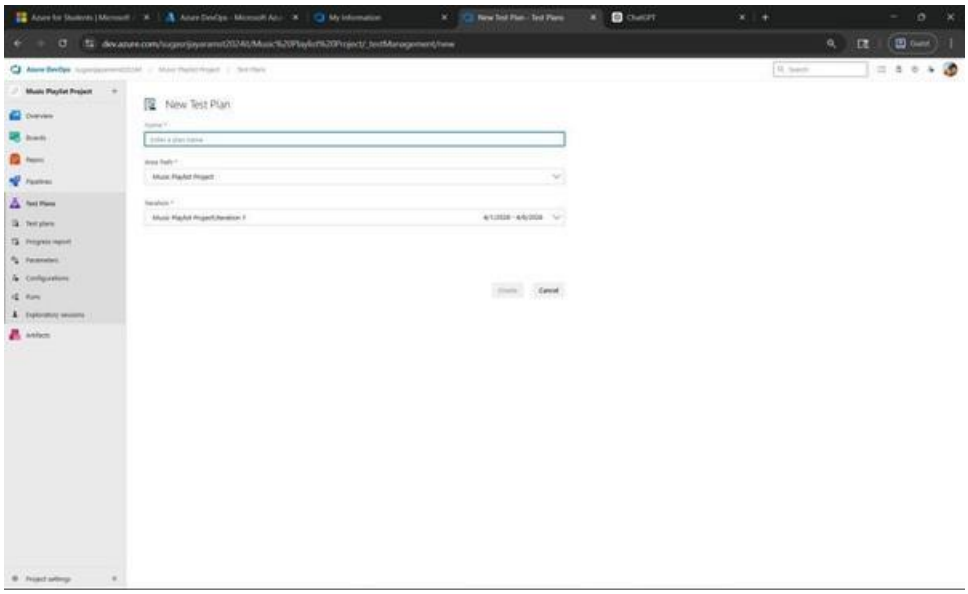
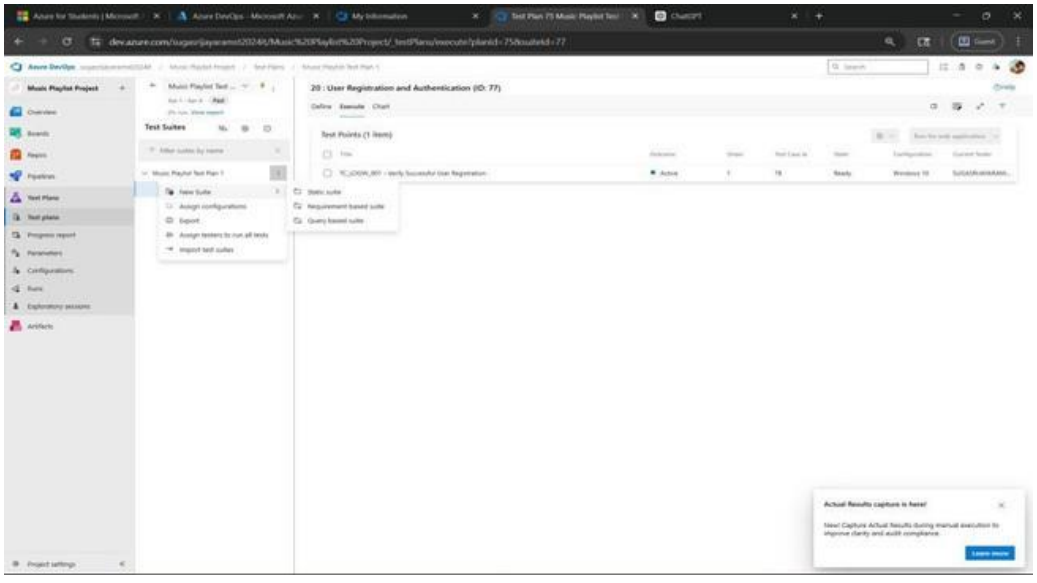
1. New test plan



2. Test suite



3. Test case

Give two test cases for at least two user stories showcasing the happy path and error scenarios in azure DevOps platform.
Music Playlist Project – Test Plans

USER STORIES

- As a user, I want to register and log in securely so that I can access my personal music playlists and downloads. (ID: 20).
- As a user, I want to create and manage playlists so that I can organize my favourite songs easily (ID: 23).

Test Suites

Test Suit: TS01 - User Registration and Authentication (ID: 20)

---

1. TC01 – User Registration Successfully w/o Error

- **Action:**
  - Go tothe Sign-Uppage
  - Entervalid name,email, and password
  - Click"Sign Up"
- **Expected Results:**
  - Sign-Up form isdisplayed
  - Fields accept valueswithout error
  - Account is createdsuccessfully
  - User is redirectedtodashboard
- **Type: Happy Path**

2. TC02 – User Registration Failed with Invalid User Details

- **Action:**
  - Go tothe Sign-Up page
  - Enterinvalid details (empty fields / wrong email format / weak password)
  - Click"Sign Up"
- **Expected Results:**
  - Validation errorsare displayed
  - Account is not created
- **Type: Error Path**

3. TC03 – User Login Successfully with Valid Credentials

- **Action:**
  - Go toLogin page
  - Entervalid email and password
  - Click"Login"
- **Expected Results:**
  - Login form isdisplayed
  - Credentials areaccepted
  - User is redirectedto dashboard
- **Type: Happy Path**

4. TC04 – Login with Invalid Credentials

- **Action:**
  - Go toLogin page
  - Enterinvalid email or password
  - Click"Login"
- **Expected Results:**
  - Input is accepted
  - Error message "Invalid username or password" is shown
- **Type: Error Path**

Test Suit: TS02 - Playlist Creation and Management (ID: 23)

---

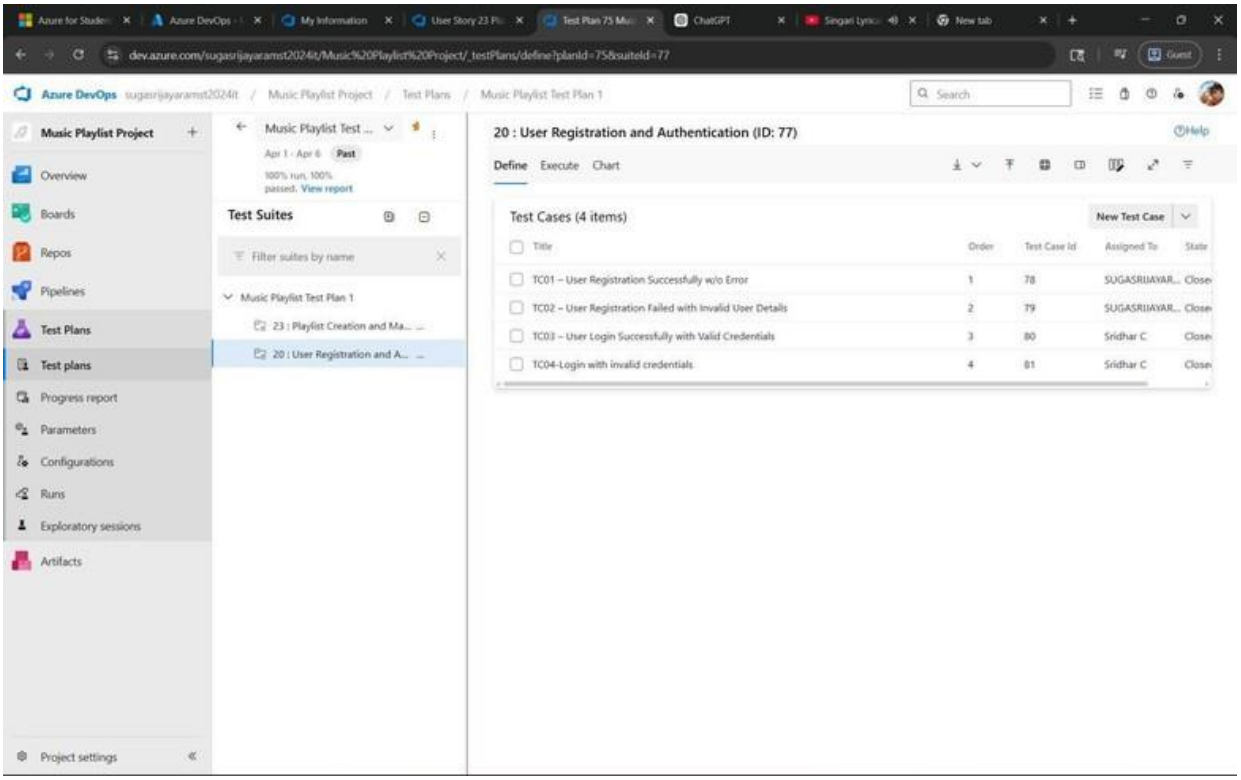
1. **TC05 – Create Playlist Successfully**
   - **Action:**
     - Login with valid credentials
     - Navigate to "My Playlists"
     - Click "Create Playlist"
     - Enter playlist name
     - Click "Save/Create"
   - **Expected Results:**
     - Playlist is created successfully
     - Playlist appears in list
   - **Type:** Happy Path

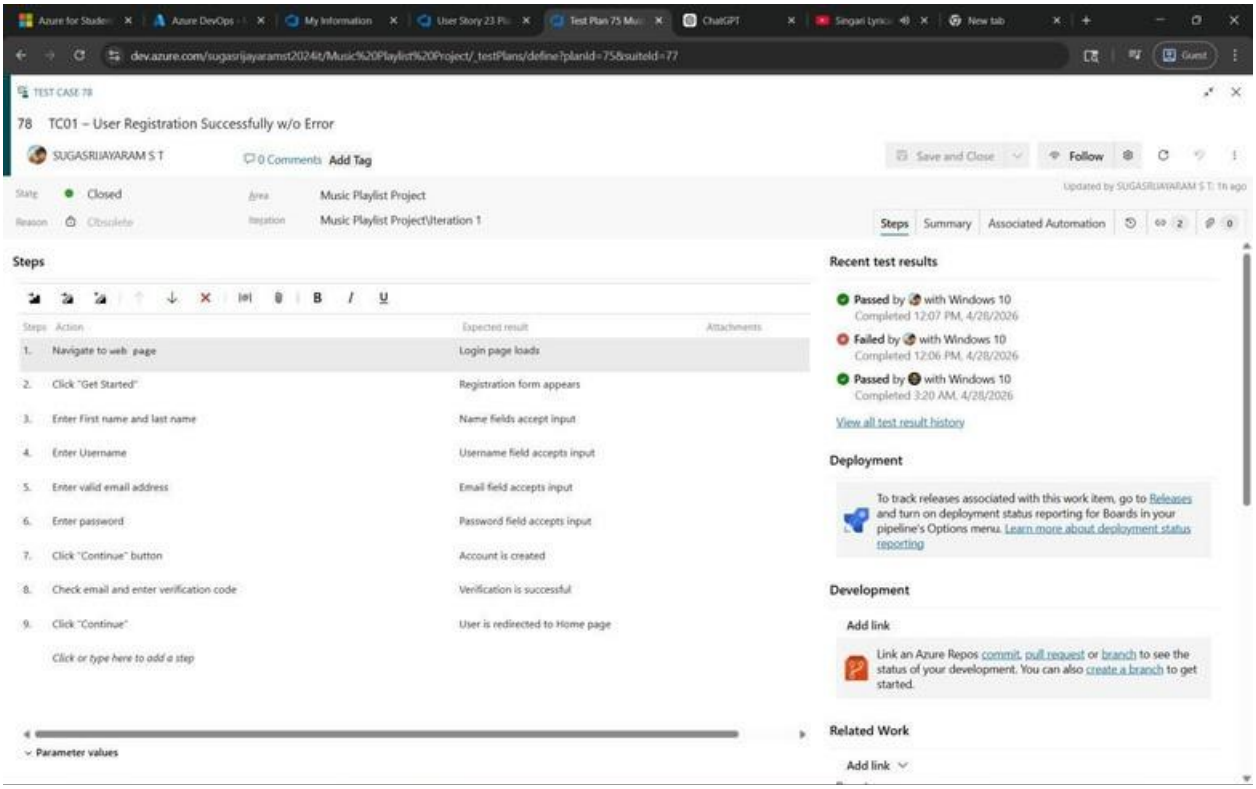
2. **TC06 – Add and Remove Songs from Playlist**
    - **Action:**
        - Open anexisting playlist
        - Click "Add Song" and select a song
        - Click "Remove" on a song
    - **Expected Results:**
        - Song is added to playlist
        - Song is removed from playlist
    - **Type:** Happy Path

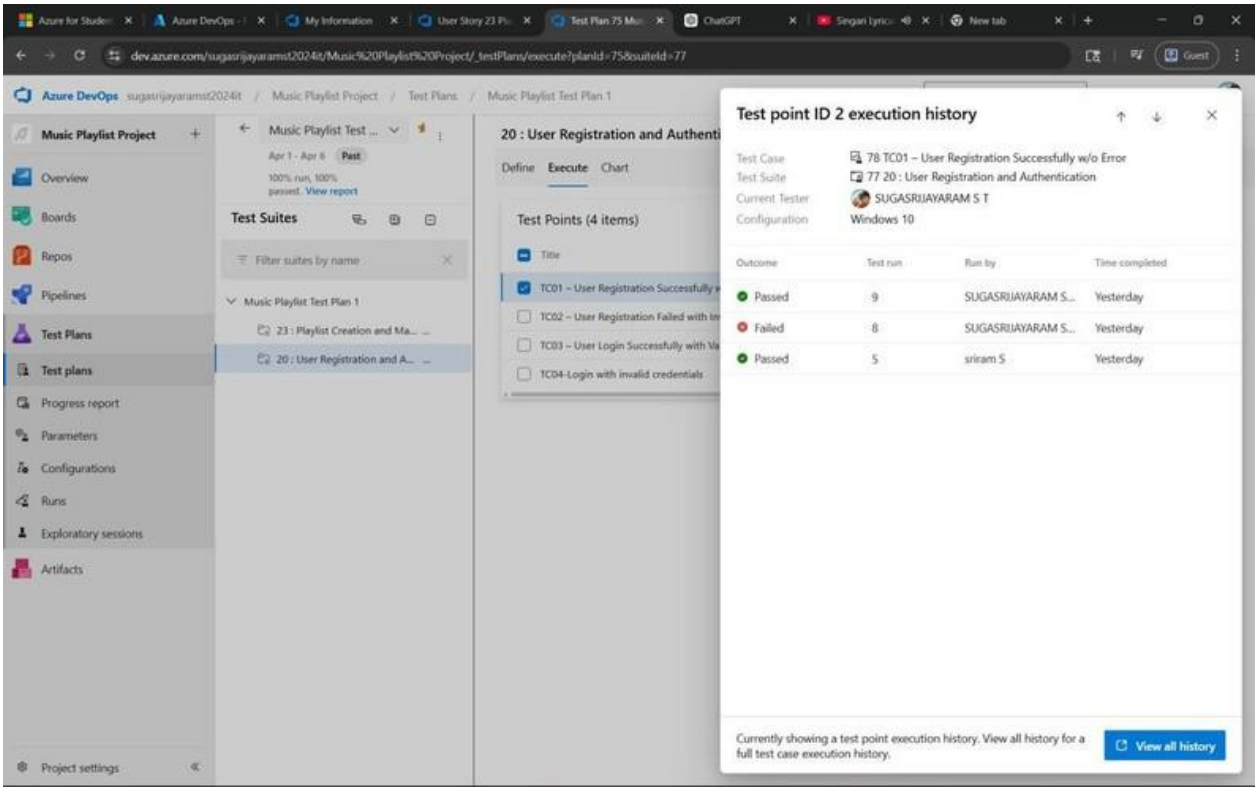
3. **TC08 – Delete Playlist and Save Changes**
    - **Action:**
        - Navigate to "My Playlists"
        - Click "Delete" on a playlist
        - Confirm deletion
    - **Expected Results:**
        - Playlist is deleted successfully
        - Playlist is removed from list
        - Changes are saved

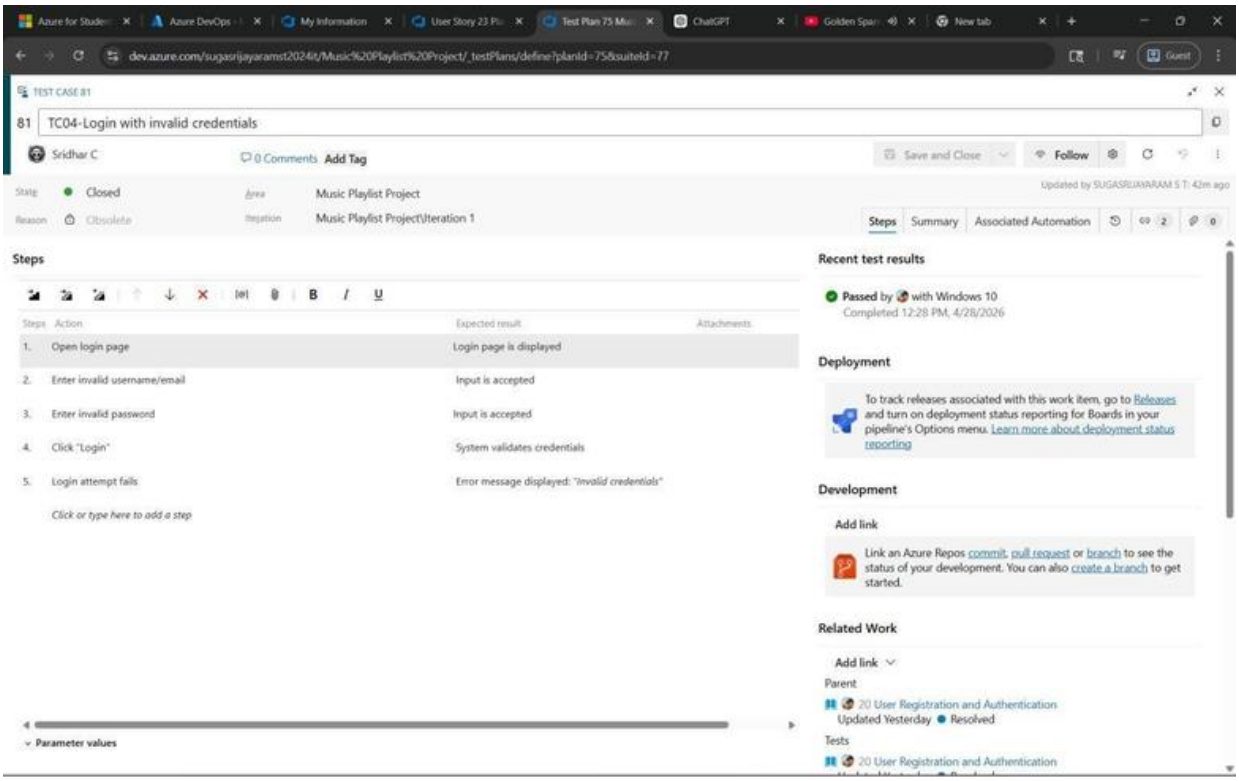
    Type: Happy Path

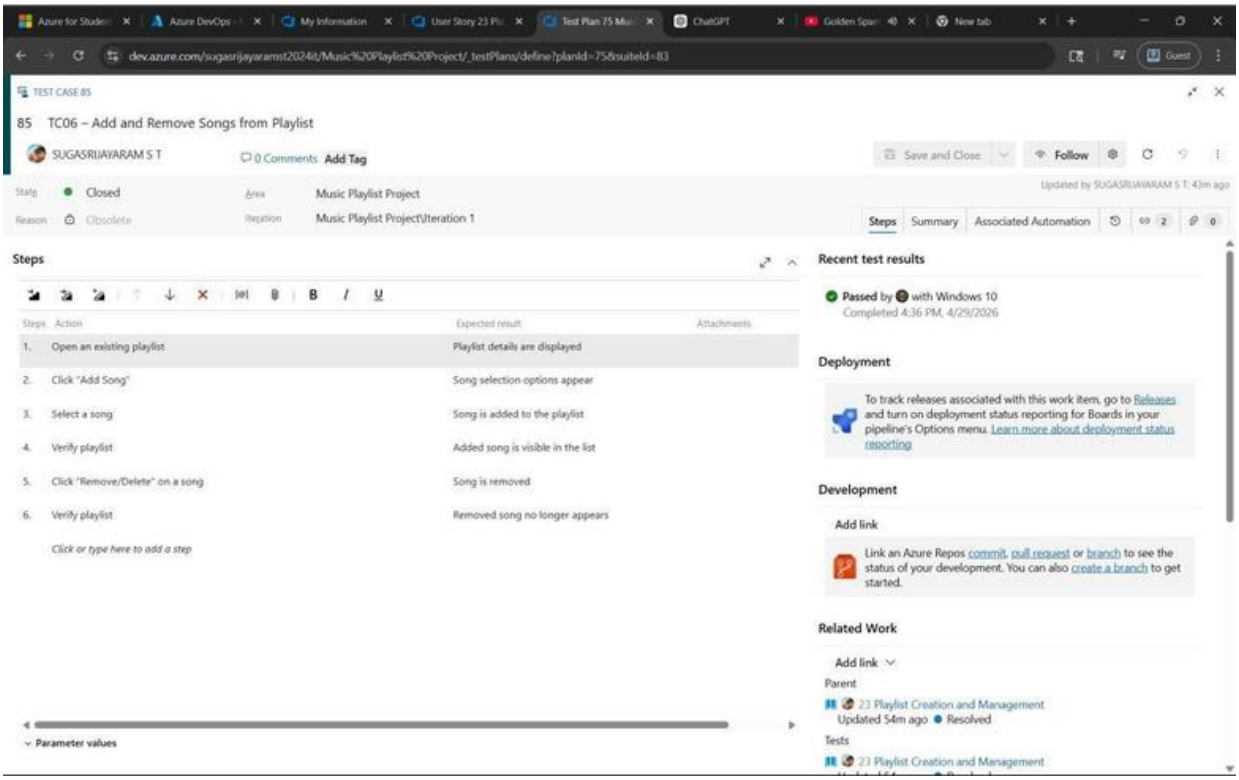
Test Cases



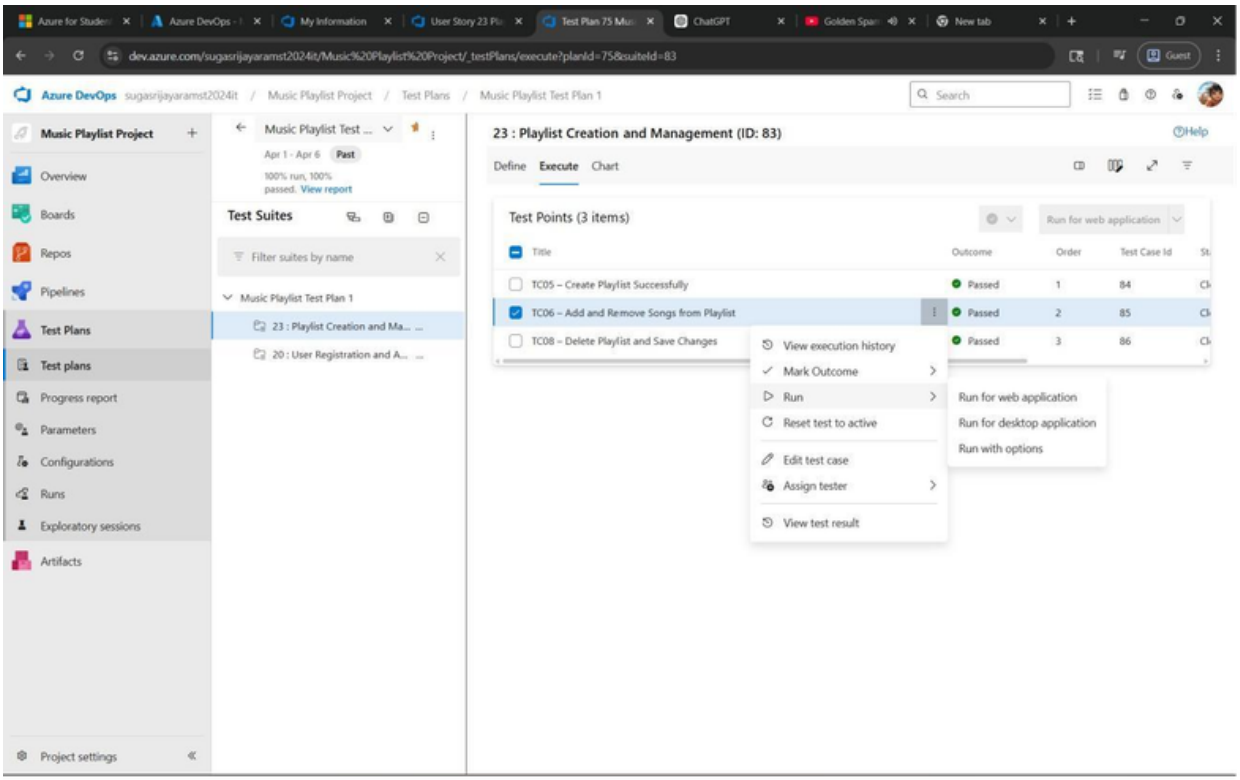

TEST CASE 78
78 TC01 – User Registration Successfully w/o Error
SUGASRIJAYARAM S T
0 Comments Add Tag
Save and Close
Follow
State Closed
Reason Obsolete
Area Music Playlist Project
Iteration Music Playlist Project\Iteration 1
Steps Summary Associated Automation
Steps
Steps Action Expected result Attachments
1. Navigate to web page Login page loads
2. Click "Get Started" Registration form appears
3. Enter First name and last name Name fields accept input
4. Enter Username Username field accepts input
5. Enter valid email address Email field accepts input
6. Enter password Password field accepts input
7. Click "Continue" button Account is created
8. Check email and enter verification code Verification is successful
9. Click "Continue" User is redirected to Home page
Click or type here to add a step
Parameter values
Recent test results
Passed by with Windows 10
Completed 12:07 PM, 4/28/2026
Failed by with Windows 10
Completed 12:06 PM, 4/28/2026
Passed by with Windows 10
Completed 3:20 AM, 4/28/2026
View all test result history
Deployment
To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting
Development
Add link
Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.
Related Work
Add link

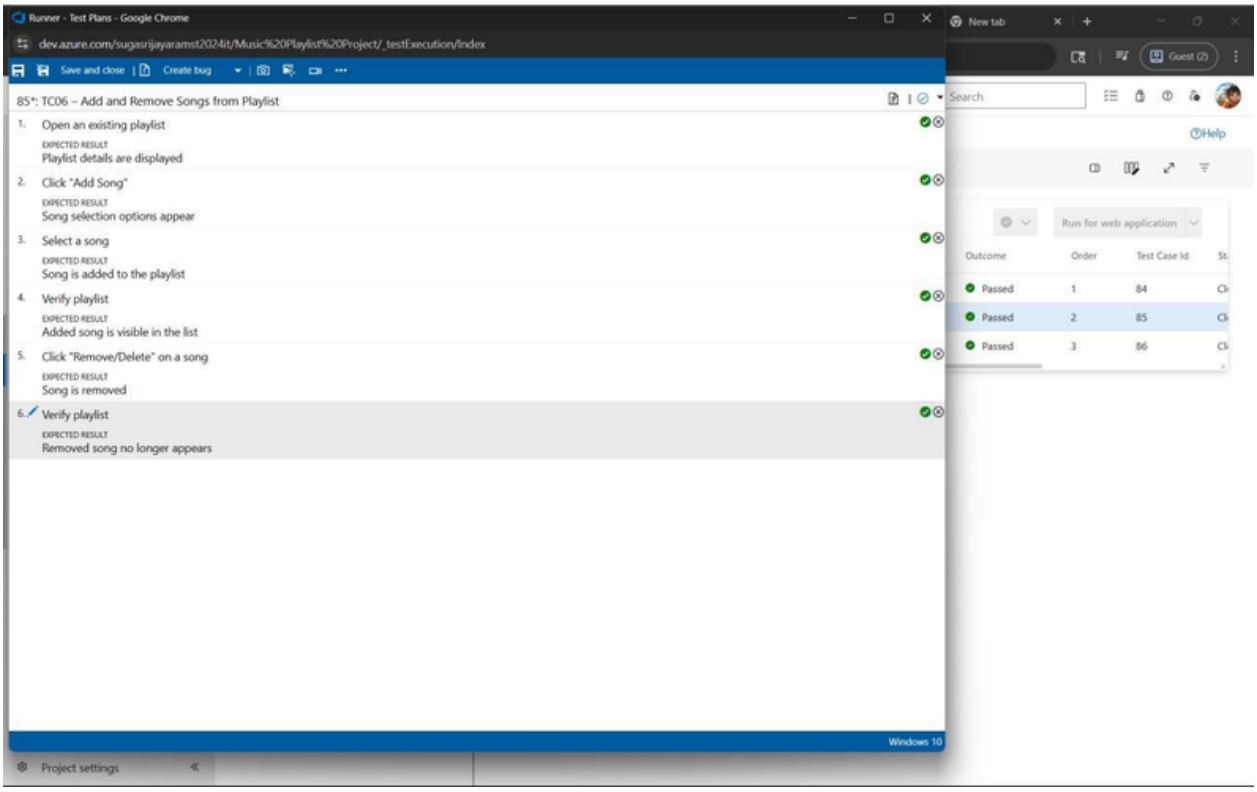

Azure DevOps sugasrijayaramst2024it / Music Playlist Project / Test Plans / Music Playlist Test Plan 1
Music Playlist Project
Overview
Boards
Repos
Pipelines
Test Plans
Test plans
Progress report
Parameters
Configurations
Runs
Exploratory sessions
Artifacts
Project settings
Test Suites
Filter suites by name
Music Playlist Test Plan 1
23 : Playlist Creation and Ma...
20 : User Registration and A...
20 : User Registration and Authenti
Define Execute Chart
Test Points (4 items)
Title
TC01 – User Registration Successfully
TC02 – User Registration Failed with
TC03 – User Login Successfully with
TC04-Login with invalid credentials
Test point ID 2 execution history
Test Case 78 TC01 – User Registration Successfully w/o Error
Test Suite 77 20 : User Registration and Authentication
Current Tester SUGASRIJAYARAM S T
Configuration Windows 10
Outcome Test run Run by Time completed
Passed 9 SUGASRIJAYARAM S... Yesterday
Failed 8 SUGASRIJAYARAM S... Yesterday
Passed 5 sriram S Yesterday
Currently showing a test point execution history. View all history for a full test case execution history.
View all history


dev.azure.com/sugasrijayaramst2024it/Music%20Playlist%20Project/_testPlans/define?planId=75&suiteId=77
TEST CASE 81
81 TC04-Login with invalid credentials
Sridhar C
0 Comments Add Tag
Save and Close
Follow
State Closed
Area Music Playlist Project
Reason Obsolete
Iteration Music Playlist Project\Iteration 1
Updated by SUGASRIJAYARAM S T: 42m ago
Steps Summary Associated Automation
Steps
Steps Action Expected result Attachments
1. Open login page Login page is displayed
2. Enter invalid username/email Input is accepted
3. Enter invalid password Input is accepted
4. Click "Login" System validates credentials
5. Login attempt fails Error message displayed: "Invalid credentials"
Click or type here to add a step
Parameter values
Recent test results
Passed by with Windows 10
Completed 12:28 PM, 4/28/2026
Deployment
To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting
Development
Add link
Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.
Related Work
Add link
Parent
20 User Registration and Authentication
Updated Yesterday Resolved
Tests
20 User Registration and Authentication

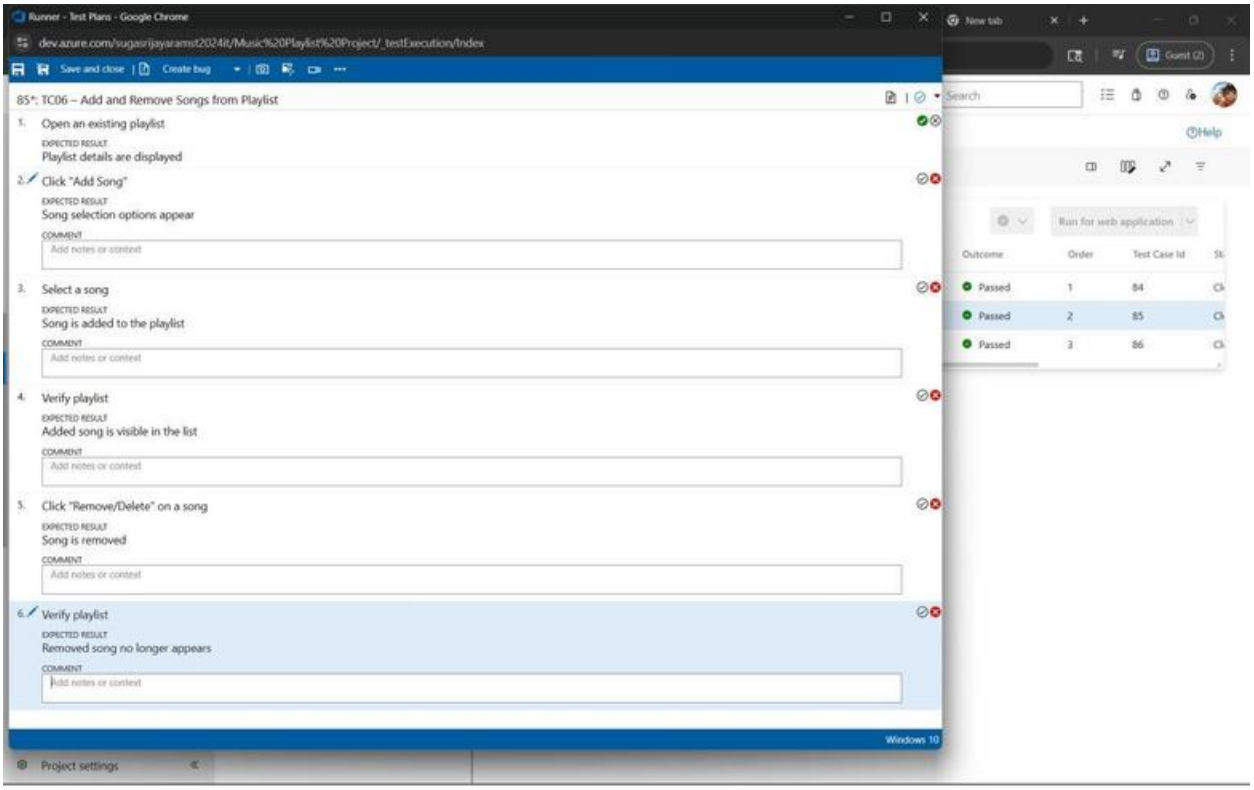

dev.azure.com/sugasrijayaramst2024it/Music%20Playlist%20Project/_testPlans/define?planId=75&suiteId=83
TEST CASE 85
85 TC06 – Add and Remove Songs from Playlist
SUGASRIJAYARAM S T
0 Comments Add Tag
Save and Close
Follow
State Closed
Area Music Playlist Project
Reason Obsolete
Iteration Music Playlist Project\Iteration 1
Updated by SUGASRIJAYARAM S T: 43m ago
Steps Summary Associated Automation
Steps
Steps Action Expected result Attachments
1. Open an existing playlist Playlist details are displayed
2. Click "Add Song" Song selection options appear
3. Select a song Song is added to the playlist
4. Verify playlist Added song is visible in the list
5. Click "Remove/Delete" on a song Song is removed
6. Verify playlist Removed song no longer appears
Click or type here to add a step
Parameter values
Recent test results
Passed by with Windows 10
Completed 4:36 PM, 4/29/2026
Deployment
To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting
Development
Add link
Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.
Related Work
Add link
Parent
23 Playlist Creation and Management
Updated 54m ago Resolved
Tests
23 Playlist Creation and Management

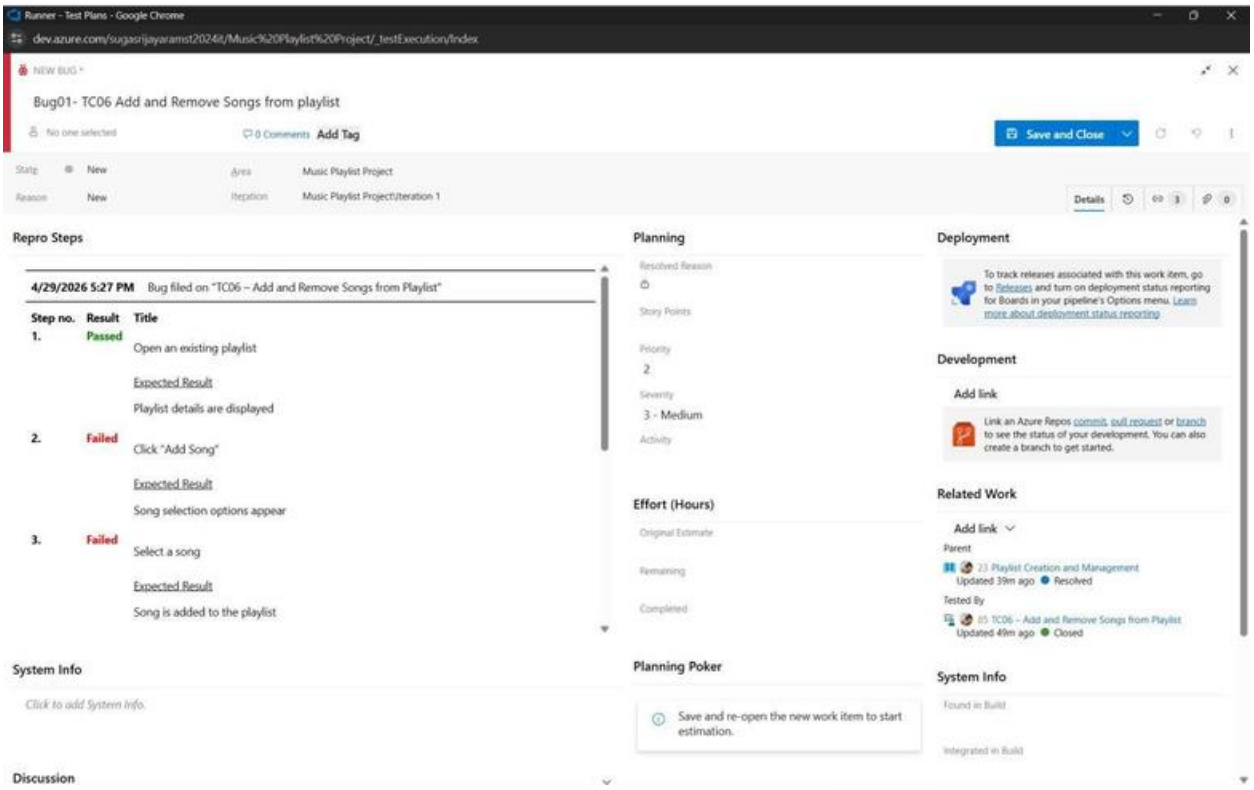
4. Running the test cases



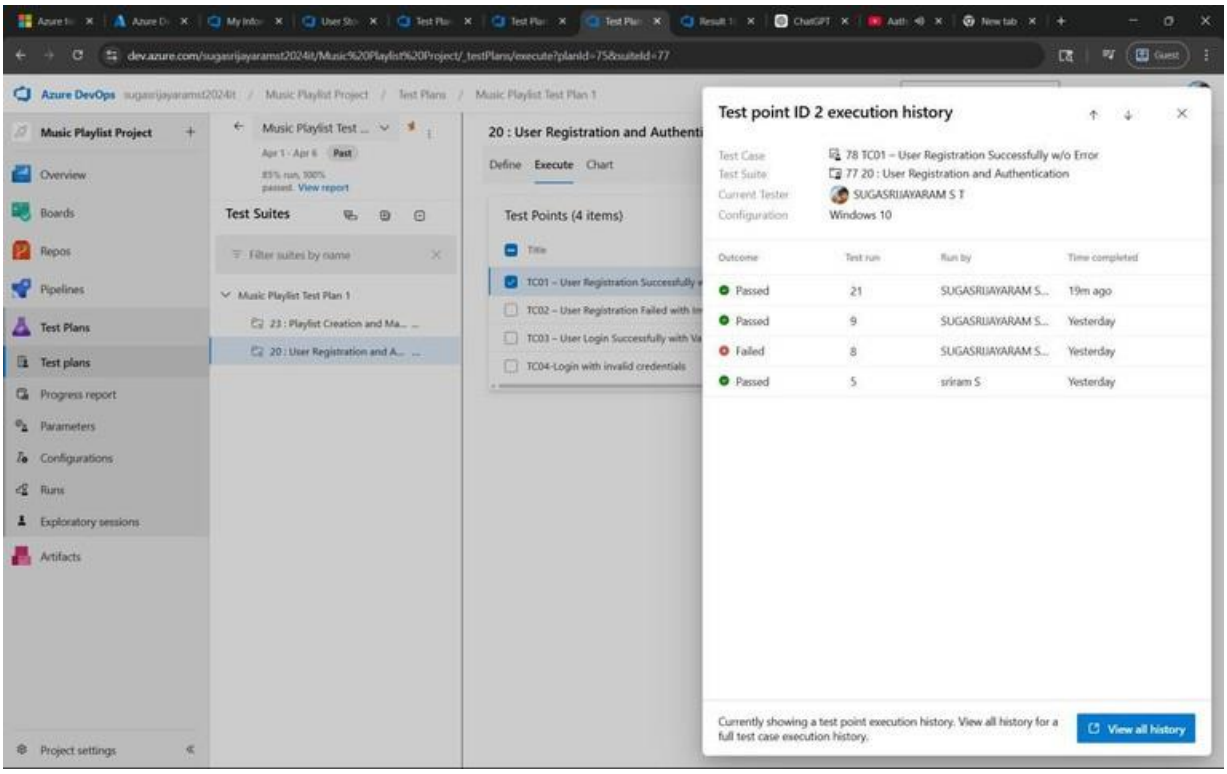


5. Creating the bug



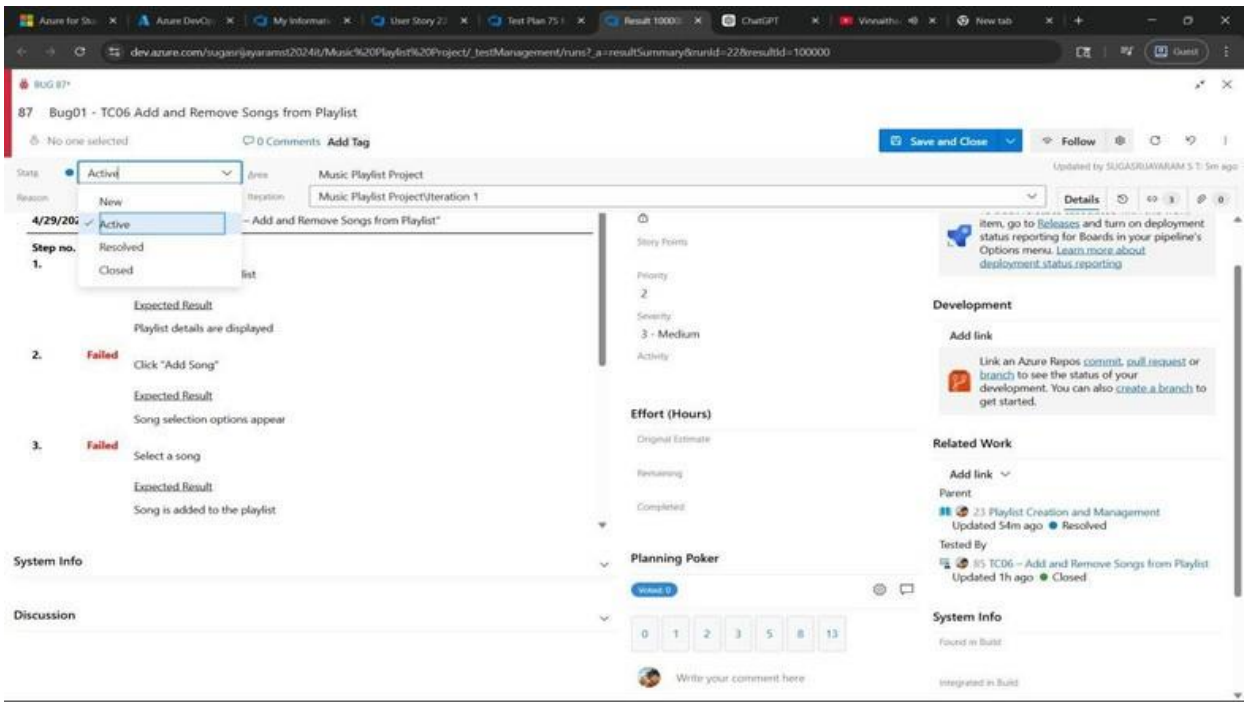


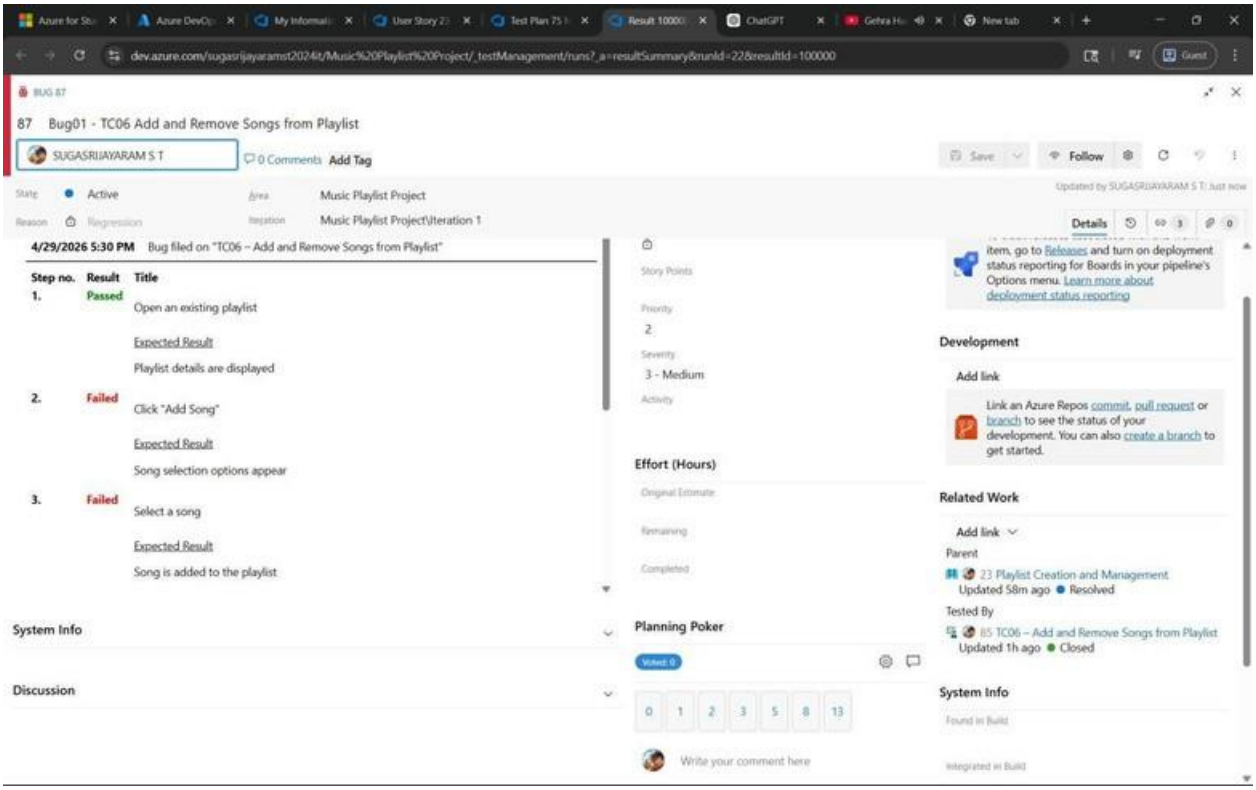
6. Test case results



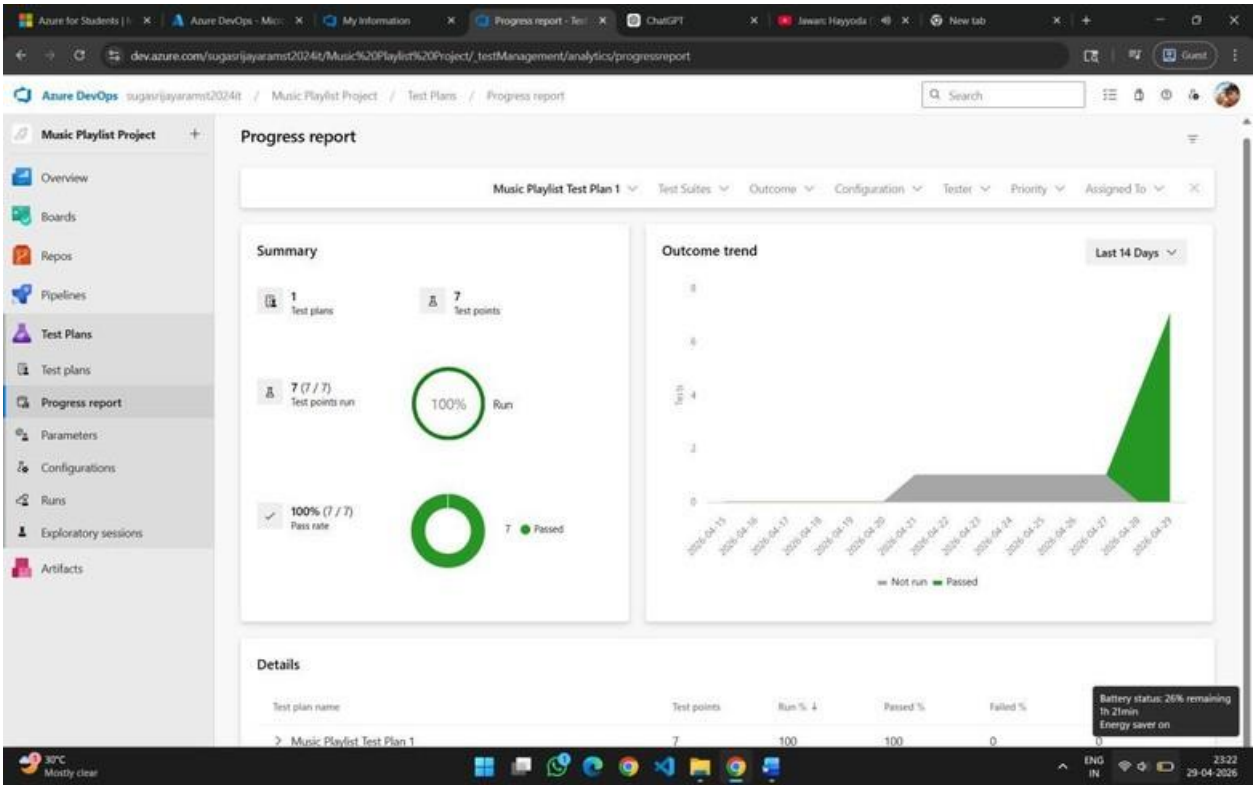
7. Bug report

- Assigning bug to the developer and changing state to Active





8. Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path.

EXPNO: 9 Create Epic,Features, User Stories, Task	LOAD TESTING AND PIPELINES

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
   Go to https://portal.azure.com and log in.
2. Create the Resource
   - Go to *Create a resource* → Search for "Azure Load Testing".
   - Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
   - Subscription*:* Choose your Azure subscription.
   - *Resource Group:* Create new or select an existing one.
   - *Name:* Provide a unique name (no special characters).
   - *Location:* Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

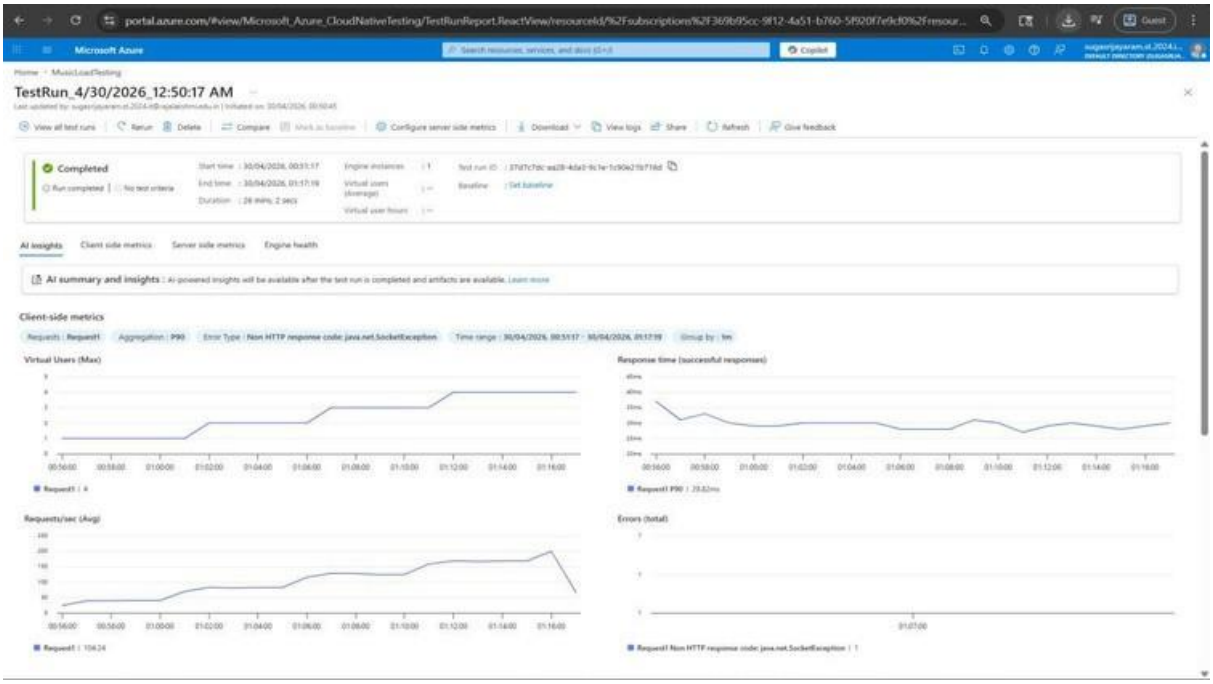
Steps to Create and Run a Load Test:

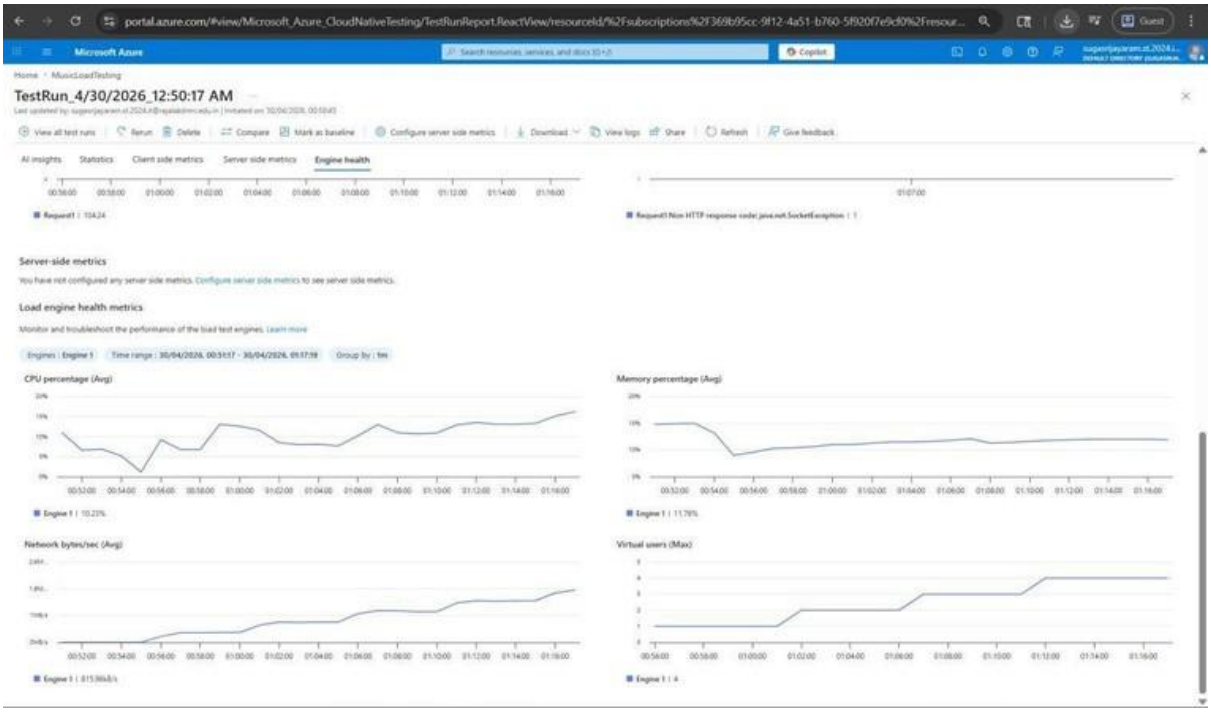
Once your resource is ready:

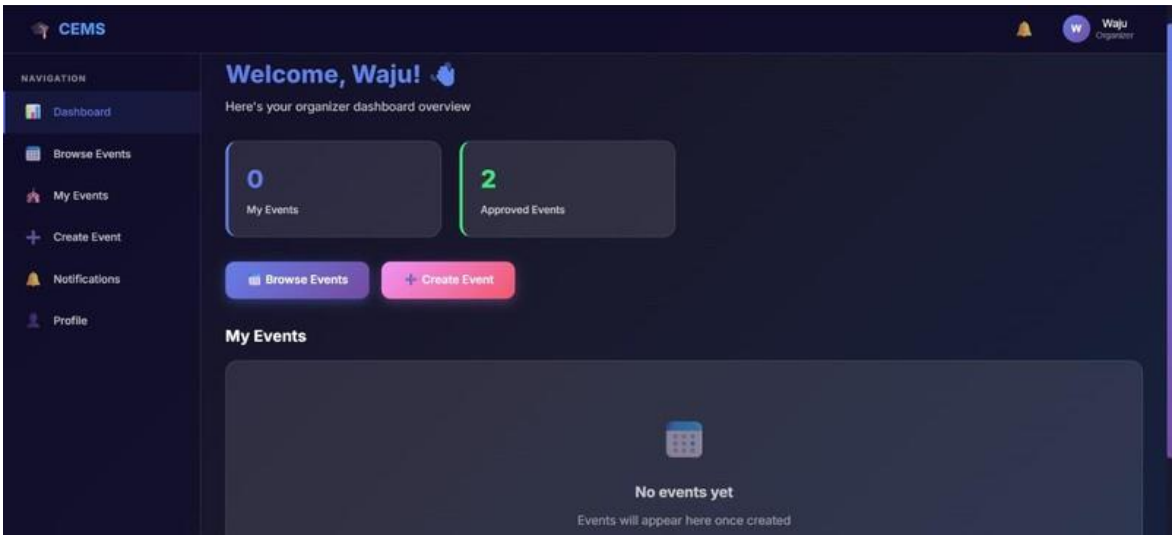
1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
   - *Test Name:* Provide a unique name.
   - *Description:* (Optional) Add test purpose.
   - *Run After Creation:* Keep checked.
3. Load Settings
   - *Test URL:* Enter the target endpoint (e.g., https://yourapi.com/products).

4. Click Review + Create → Create to start the test.

Load Testing







Pipelines

Description:

Thisexperiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
    - In Azure DevOps, create a new project.
    - Create a pipeline and select GitHub as the source.
    - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
    - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
    - Add the following basic pipeline configuration for Python and Flask:

yml Code

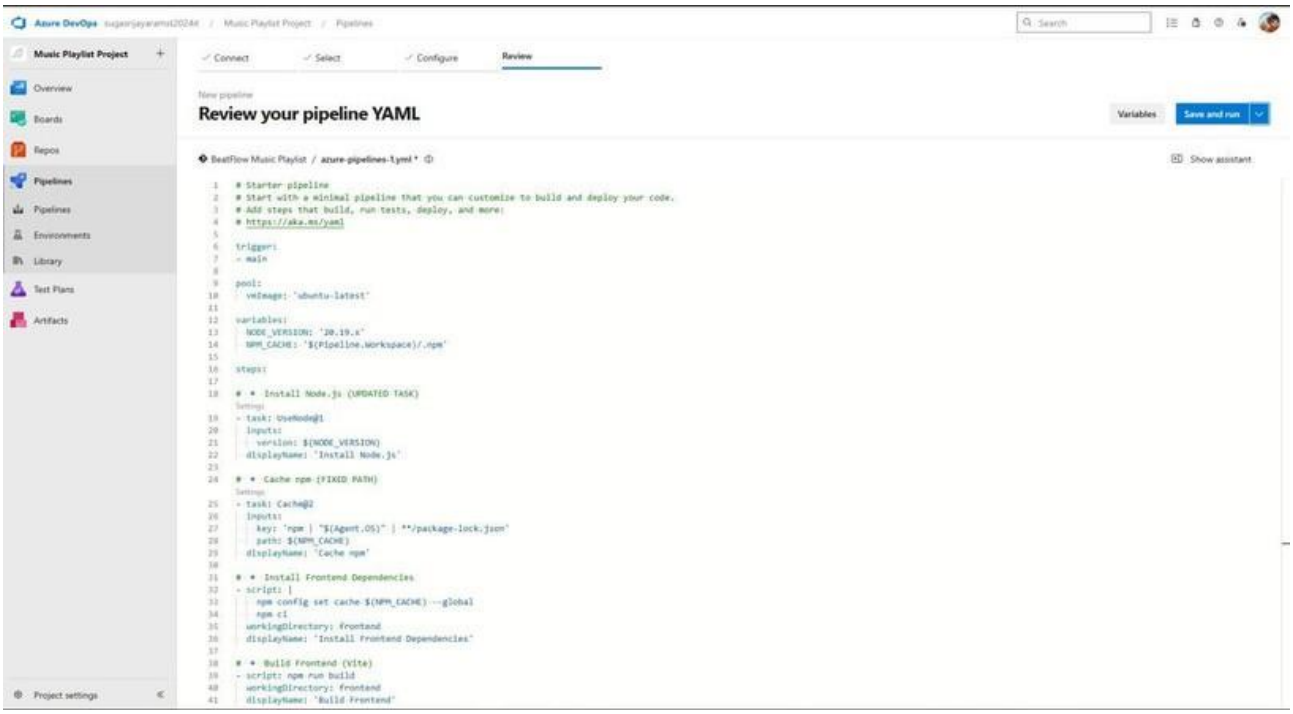
```
trigger:
- main

pool:
  vmImage: 'ubuntu-latest'

variables:
  NODE_VERSION: '20.19.x'
  NPM_CACHE: '$(Pipeline.Workspace)/.npm'

steps:

# 🔹 Install Node.js (UPDATED TASK)
- task: UseNode@1
  inputs:
    version: $(NODE_VERSION)
  displayName: 'Install Node.js'

# 🔹 Cache npm (FIXED PATH)
- task: Cache@2
  inputs:
    key: 'npm | "$(Agent.OS)" | **/package-lock.json'
    path: $(NPM_CACHE)
  displayName: 'Cache npm'

# 🔹 Install Frontend Dependencies
- script: |
    npm config set cache $(NPM_CACHE) --global
    npm ci
  workingDirectory: frontend
  displayName: 'Install Frontend Dependencies'

# 🔹 Build Frontend (Vite)
- script: npm run build
  workingDirectory: frontend
  displayName: 'Build Frontend'

# 🔹 Frontend Tests (optional)
```

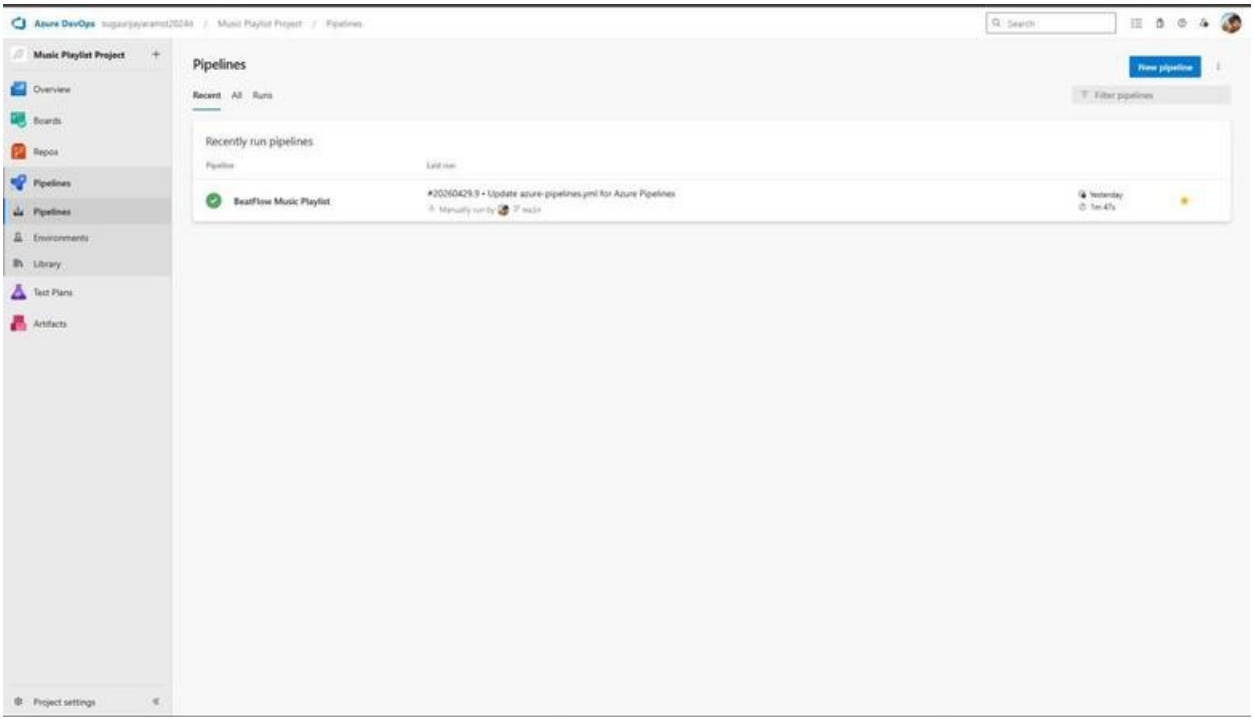
```
- script: npm test --if-present
  workingDirectory: frontend
  displayName: 'Run Frontend Tests'

# 🔹 Install Backend Dependencies
- script: |
    npm config set cache $(NPM_CACHE) --global
    npm ci
  workingDirectory: backend
  displayName: 'Install Backend Dependencies'

# 🔹 Backend Tests
- script: npm test --if-present
  workingDirectory: backend
  displayName: 'Run Backend Tests'

# 🔹 Verify Frontend Build (fail early)
- script: |
    if [ ! -d "frontend/dist" ]; then
      echo "Build failed: dist folder not found"
      exit 1
    fi
  displayName: 'Verify Frontend Build'

# 🔹 Archive Frontend
- task: ArchiveFiles@2
  inputs:
    rootFolderOrFile: 'frontend/dist'
    includeRootFolder: false
    archiveType: 'zip'
    archiveFile: '$(Build.ArtifactStagingDirectory)/frontend.zip'
  displayName: 'Archive Frontend'

# 🔹 Archive Backend
- task: ArchiveFiles@2
  inputs:
    rootFolderOrFile: 'backend'
    includeRootFolder: false
    archiveType: 'zip'
    archiveFile: '$(Build.ArtifactStagingDirectory)/backend.zip'
  displayName: 'Archive Backend'

# 🔹 Publish Artifacts
```

```
- task: PublishBuildArtifacts@1
  inputs:
    PathtoPublish: '$(Build.ArtifactStagingDirectory)'
    ArtifactName: 'drop'
  displayName: 'Publish Artifacts'
```

3. Run and Monitor Pipeline:
   - Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
   - Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline





Result:

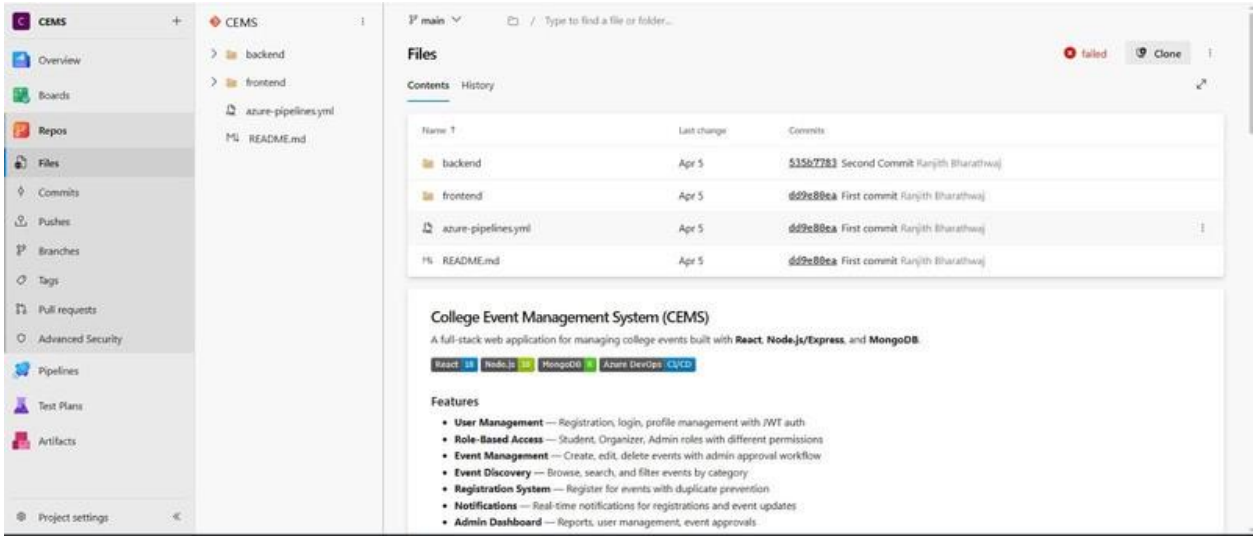
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

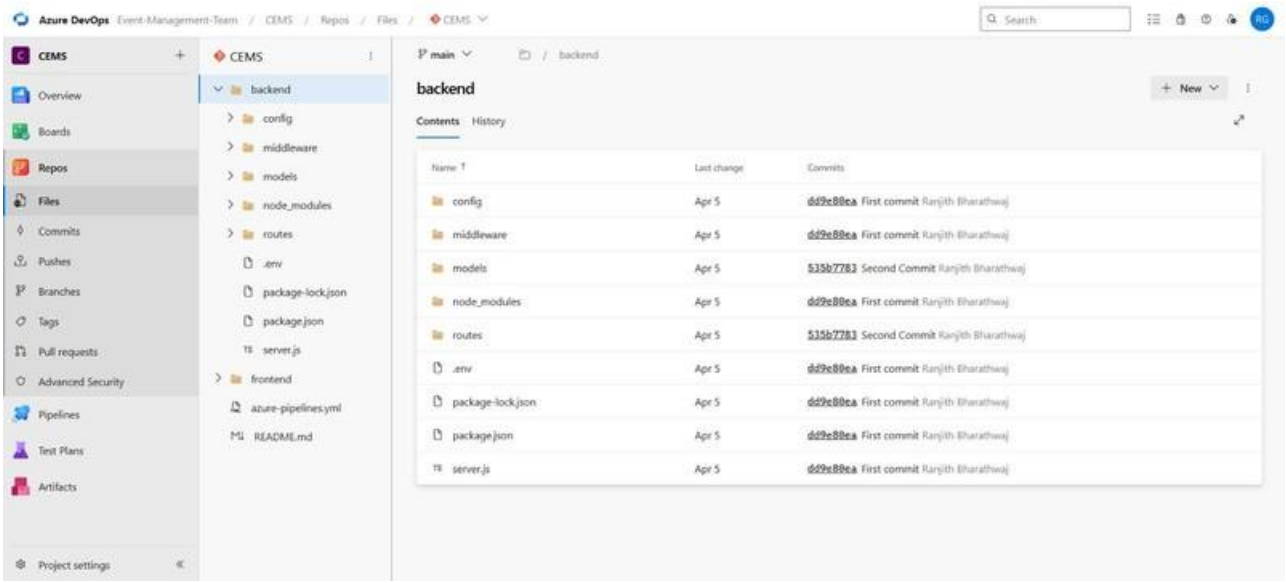
EXPNO: 10 Create Epic, Features, User Stories, Task	GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the College Event Management.

GitHub Project Structure





Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.